Unit 1: Numerical Expressions & Factors — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 1 • Manipulative Kits: Counting & Array Studio (COUNT-01) + Prime Factor Studio (PRIME-01)

Target Misconception: Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).

Hands-On Block Remediation: Have students build physical tile rows for division remainders, stack 2D/3D arrays for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 1.1 (pp.2-3): WHOLE NUMBER OPERATIONS & DIVISION WITH REMAINDERS [COUNT-01]

Methods: Method 1: Equal-Rows Array Model | Method 2: Division Algorithm & Multiplicatio..

PRACTICE A (p.2): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 5 R 1 Keys: 31 [ENTER] 6 [g][÷R]	**A2: 9 R 2** Keys: 47 [ENTER] 5 [g][÷R]
A3: 8 R 2 Keys: 58 [ENTER] 7 [g][÷R]	**A4: 9 R 4** Keys: 85 [ENTER] 9 [g][÷R]
A5: 12 R 2 Keys: 146 [ENTER] 12 [g][÷R]	**A6: 6 R 34** Keys: 250 [ENTER] 36 [g][÷R]
A7: 117 Keys: 14 [ENTER] 8 [×] 5 [+]	**A8: 9 R 3 (Rem < 8)** Keys: 75 [ENTER] 8 [g][÷R]
A9: 7 boxes, 6g left (Way 2: 7×42=294) Keys: 300 [ENTER] 42 [g][÷R]	**A10: 9 vans (8 full + 1 for 3 kids; Way 2:..** Keys: 115 [ENTER] 14 [g][÷R]

PRACTICE B (p.3): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 8 R 5 (8 full packs, 5 cells left) Keys: 53 [ENTER] 6 [g][÷R]	**B2: 11 R 6 (11 figures, 6 tickets left)** Keys: 94 [ENTER] 8 [g][÷R]
B3: 19 R 2 -> 20 canoes Keys: 78 [ENTER] 4 [g][÷R]	**B4: 9 R 2 (9 full racks, 2 on last)** Keys: 110 [ENTER] 12 [g][÷R]
B5: 28 R 2 (28 knights, 2 m left) Keys: 450 [ENTER] 16 [g][÷R]	**B6: 18 R 3 -> 18 full, 19 pages total** Keys: 165 [ENTER] 9 [g][÷R]
B7: 187 marbles Keys: 12 [ENTER] 15 [×] 7 [+]	**B8: 8 R 8 (8 full trains, 8 on final)** Keys: 200 [ENTER] 24 [g][÷R]
B9: 8 R 3 -> 9 pizzas (5 slices left) Keys: 67 [ENTER] 8 [g][÷R]	**B10: 11 R 17 (11×28 + 17 = 325)** Keys: 325 [ENTER] 28 [g][÷R]

SEC 1.2 (pp.4-5): POWERS AND EXPONENTS [COUNT-01]

Methods: Method 1: Repeated Multiplication Factors | Method 2: 2D Square / 3D Cube Array..

PRACTICE A (p.4): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 25 Keys: 5 [x²]	**A2: 49 (vs 14)** Keys: 7 [x²]
A3: 27 Keys: 3 [g][x³]	**A4: 125** Keys: 5 [g][x³]
A5: 32 Keys: 2 [ENTER] 5 [g][y^x]	**A6: 1,296** Keys: 6 [ENTER] 4 [g][y^x]
A7: 10,000 and 1 Keys: 10 [ENTER] 4 [g][y^x]	**A8: 16 (2×2×2×2)** Keys: 2 [ENTER] 4 [g][y^x]
A9: 64 cells (Way 2: 2 [g][REP] [×]×5) Keys: 2 [ENTER] 6 [g][y^x]	**A10: 6 blocks wide (Way 2: 6 [g][x²]=216)** Keys: 216 [ENTER] 3 [g][x√y]

PRACTICE B (p.5): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 8 × 8 = 64 pixels Keys: 8 [x²]	**B2: 216 blocks** Keys: 6 [g][x³]
B3: 64 bacteria Keys: 2 [ENTER] 6 [g][y^x]	**B4: 256 players** Keys: 4 [ENTER] 4 [g][y^x]
B5: 144 + 25 = 169 ft² Keys: 12 [x²] 5 [x²] [+]	**B6: 625 students** Keys: 5 [ENTER] 4 [g][y^x]
B7: 14 meters Keys: 196 [f][√x]	**B8: 2^5 = 32 > 5² = 25 (diff = 7)** Keys: 2 [ENTER] 5 [g][y^x] 5 [x²] [-]
B9: 1,024 bytes Keys: 2 [ENTER] 10 [g][y^x]	**B10: 27 + 343 = 370 in³** Keys: 3 [g][x³] 7 [g][x³] [+]

SEC 1.3 (pp.6-7): ORDER OF OPERATIONS [COUNT-01]

Methods: Method 1: PEMDAS Line-by-Line Funnel | Method 2: Inside-Out Grouping Chunks | E..

PRACTICE A (p.6): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 11 Keys: 5 [ENTER] 2 [ENTER] 3 [×] [+]	**A2: 4** Keys: 14 [ENTER] 6 [+] 7 [ENTER] 2 [-] [÷]
A3: 6 Keys: 30 [ENTER] 4 [ENTER] 6 [×] [-]	**A4: 84** Keys: 9 [ENTER] 3 [+] 15 [ENTER] 8 [-] [×]
A5: 1 Keys: 50 [ENTER] 3 [ENTER] 4 [+] [x²] [-]	**A6: 20** Keys: 120 [ENTER] 4 [ENTER] 3 [ENTER] 2 [+] [x²]..
A7: 6 Keys: 24 [ENTER] 12 [+] 9 [ENTER] 3 [-] [÷]	**A8: 24** Keys: 3 [g][x³] 15 [ENTER] 3 [÷] [+] 2 [ENTER] 4..
A9: 26 (Way 2: 10 + 144÷9 = 26) Keys: 10 [ENTER] 15 [ENTER] 3 [-] [x²] 9 [÷] [+]	**A10: $46 (Way 2: 24 + 32 - 10 = 46)** Keys: 2 [ENTER] 12 [×] 4 [ENTER] 8 [×] [+] 10 [-]

PRACTICE B (p.7): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: $56 Keys: 9 [ENTER] 5 [+] 4 [×]	**B2: $73** Keys: 45 [ENTER] 4 [ENTER] 7 [×] [+]
B3: 101 points Keys: 100 [ENTER] 3 [ENTER] 8 [×] [-] 25 [+]	**B4: 64 square units** Keys: 10 [x²] 6 [x²] [-]
B5: $73 Keys: 2 [ENTER] 14 [×] 5 [ENTER] 9 [×] [+]	**B6: 70 points** Keys: 3 [x²] 5 [+] 6 [×] 14 [-]
B7: 10 boxes Keys: 72 [ENTER] 12 [-] 6 [÷]	**B8: 42** Keys: 48 [ENTER] 6 [÷] 4 [×] 10 [+]
B9: 12 muffins per basket Keys: 5 [ENTER] 12 [×] 3 [ENTER] 8 [×] [+] 7 [÷]	**B10: 74** Keys: 4 [g][x³] 14 [-] 2 [ENTER] 3 [+] [÷] 8 [x²]..

SEC 1.4 (pp.8-9): PRIME FACTORIZATION [PRIME-01]

Methods: Method 1: Factor Tree Diagram | Method 2: Prime Division Ladder & Logarithmic P..

PRACTICE A (p.8): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 2² × 3 Keys: 12 [g][FACT]	**A2: 2² × 3²** Keys: 36 [g][FACT]
A3: 29 Prime; 57 Composite (3×19) Keys: 29 [PRIME], 57 [g][FACT]	**A4: 3² × 5** Keys: 45 [g][FACT]
A5: 2³ × 3² Keys: 72 [g][FACT]	**A6: 2² × 3 × 7** Keys: 84 [g][FACT]
A7: 2² × 5² Keys: 100 [g][FACT]	**A8: 84** Keys: 2 [x²] 3 [×] 7 [×]
A9: 2² × 3² × 5 (Way 2: (2·3²)(2·5)) Keys: 180 [g][FACT]	**A10: 360 (Way 2: 8 × 9 × 5 = 360)** Keys: 2 [g][x²] 3 [x²] [×] 5 [×]

PRACTICE B (p.9): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 2^4 × 3 Keys: 48 [g][FACT]	**B2: 2 × 3³** Keys: 54 [g][FACT]
B3: 51 is composite (3 × 17) Keys: 51 [g][FACT]	**B4: 2^4 × 5** Keys: 80 [g][FACT]
B5: 126 Keys: 2 [ENTER] 3 [x²] [×] 7 [×]	**B6: 2² × 5²** Keys: 100 [g][FACT]
B7: 2² × 5 × 7 Keys: 140 [g][FACT]	**B8: 2^4 × 3²** Keys: 144 [g][FACT]
B9: 2 × 3² × 5 Keys: 90 [g][FACT]	**B10: 210** Keys: 2 [ENTER] 3 [×] 5 [×] 7 [×]

SEC 1.5 (pp.10-11): GREATEST COMMON FACTOR (GCF) & LEAST COMMON MULTIPLE (LCM) [PRIME..

Methods: Method 1: Lists of Factors & Multiples | Method 2: Prime Factorization Venn Dia..

PRACTICE A (p.10): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: GCF = 6, LCM = 36 Keys: 12 [ENTER] 18 [GCF] / [f][LCM]	**A2: GCF = 4, LCM = 24** Keys: 8 [ENTER] 12 [GCF] / [f][LCM]
A3: GCF = 5, LCM = 75 Keys: 15 [ENTER] 25 [GCF] / [f][LCM]	**A4: GCF = 7, LCM = 70** Keys: 14 [ENTER] 35 [GCF] / [f][LCM]
A5: GCF = 8, LCM = 48 Keys: 16 [ENTER] 24 [GCF] / [f][LCM]	**A6: GCF = 14, LCM = 84** Keys: 28 [ENTER] 42 [GCF] / [f][LCM]
A7: GCF = 1, LCM = 35 Keys: 5 [ENTER] 7 [GCF] / [f][LCM]	**A8: 5 × 60 = 300 (Matches 15×20!)** Keys: 15 [ENTER] 20 [GCF] ... [×]
A9: 60 (5 packs dogs, 4 packs buns; Way 2.. Keys: 12 [ENTER] 15 [f][LCM]	**A10: 12 bags (3 red, 4 blue per bag; Way 2..** Keys: 36 [ENTER] 48 [GCF]

PRACTICE B (p.11): MODELING REAL LIFE APPLICATIONS (#1-#10)

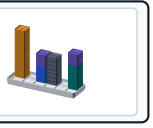
B1: 14 kits (2 bandages, 3 wipes) Keys: 28 [ENTER] 42 [GCF]	**B2: 60 seconds (1 minute)** Keys: 12 [ENTER] 15 [f][LCM]
B3: 24 (3 packs of dogs, 2 packs of buns) Keys: 8 [ENTER] 12 [f][LCM]	**B4: 16 singers per row** Keys: 32 [ENTER] 48 [GCF]
B5: 72 minutes (10:12 AM) Keys: 18 [ENTER] 24 [f][LCM]	**B6: 12 inches** Keys: 36 [ENTER] 60 [GCF]
B7: GCF = 7, LCM = 42 Keys: 14 [ENTER] 21 [GCF] 14 [ENTER] 21 [f][LCM]	**B8: 60 minutes** Keys: 4 [ENTER] 6 [f][LCM] 10 [f][LCM]
B9: 15 bouquets (3 gold, 5 silver each) Keys: 45 [ENTER] 75 [GCF]	**B10: GCF=5, LCM=60; 5×60 = 300 = 15×20!** Keys: 15 [ENTER] 20 [GCF] 15 [ENTER] 20 [f][LCM]..

Unit 2: Fractions & Decimals — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 2 • Manipulative Kits: Fraction Studio (FRAC-01) + Place Value Studio (PLACE-01)

Target Misconception: Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.

Hands-On Block Remediation: Use the 84.0 mm Fraction Studio towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 2.1 (pp.12-13): EQUIVALENT FRACTIONS, MIXED NUMBERS & LCD ADDITION [FRAC-01]

Methods: Method 1: Divide Step-by-Step by Common Factors | Method 2: Divide Once by the..

PRACTICE A (p.12): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 4/6 -> 2/3 Keys: [.] 8 [.] 12 [ENTER] [g][SIMP]×2	**A2:** 2/3 (0.667) Keys: [.] 24 [.] 36 [ENTER] [g][SIMP]...
A3: 13/5 and 2.6 Keys: 2 [.] 3 [.] 5 [ENTER] [f][d/c]	**A4:** 2 3/8 and 2.375 Keys: [.] 19 [.] 8 [ENTER] [FDISP]
A5: 5/6 (42mm+28mm=70mm) Keys: [.] 1 [.] 2 [ENTER] [.] 1 [.] 3 [+]	**A6:** 1 7/12 (or 19/12) Keys: [.] 3 [.] 4 [ENTER] [.] 5 [.] 6 [+]
A7: 5/8 Keys: [.] 7 [.] 8 [ENTER] [.] 1 [.] 4 [-]	**A8:** 2 1/2 (or 5/2) Keys: 4 [.] 1 [.] 3 [ENTER] 1 [.] 5 [.] 6 [-]
A9: 7/12 (49mm = seven 7mm twelfths) Keys: [.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [+]	**A10:** 1 7/8 miles (Way 2: 30/8 - 15/8 = 15/8) Keys: 3 [.] 3 [.] 4 [ENTER] 1 [.] 7 [.] 8 [-]

PRACTICE B (p.13): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 5/8 cup Keys: [.] 3 [.] 8 [ENTER] [.] 1 [.] 4 [+]	**B2:** 3/4 complete; 1/4 remains Keys: [.] 18 [.] 24 [g][SIMP]
B3: 13/8 and 1.625 in Keys: 1 [.] 5 [.] 8 [ENTER] [f][d/c] [FDISP]	**B4:** 11/24 mile Keys: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]
B5: 3 5/6 pizzas (23/6) Keys: 2 [.] 1 [.] 3 [ENTER] 1 [.] 1 [.] 2 [+]	**B6:** 2 3/4 feet (11/4) Keys: 4 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [-]
B7: 3/5 = 0.6 Keys: [.] 15 [.] 25 [g][SIMP] [FDISP]	**B8:** 1/5 liter (0.2 L) Keys: [.] 2 [.] 5 [ENTER] [.] 3 [.] 10 [+] [.] 1..
B9: 4 5/12 yd used; 7/12 yd left Keys: 2 [.] 2 [.] 3 [ENTER] 1 [.] 3 [.] 4 [+]	**B10:** 4 5/8 miles Keys: [.] 3 [.] 8 [ENTER] 2 [.] 1 [.] 2 [+] 1 [.]..

SEC 2.2 (pp.14-15): MULTIPLYING FRACTIONS AND MIXED NUMBERS [FRAC-01]

Methods: Method 1: Multiply Straight Across & Simplify | Method 2: Cross-Cancel Common F..

PRACTICE A (p.14): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 1/3 Keys: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×]	**A2:** 1/2 Keys: [.] 2 [.] 3 [ENTER] [.] 3 [.] 4 [×]
A3: 1/3 Keys: [.] 3 [.] 5 [ENTER] [.] 5 [.] 9 [×]	**A4:** 9 Keys: 12 [ENTER] [.] 3 [.] 4 [×]
A5: 4 Keys: 1 [.] 3 [.] 4 [ENTER] 2 [.] 2 [.] 7 [×]	**A6:** 3 Keys: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]
A7: 8 3/4 (8.75) Keys: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 2 [×]	**A8:** 3 1/3 (Less than 5!) Keys: 5 [ENTER] [.] 2 [.] 3 [×]
A9: 4 3/8 cups (Way 2: 1.75 × 2.5 = 4.375) Keys: 1 [.] 3 [.] 4 [ENTER] 2 [.] 1 [.] 2 [×]	**A10:** 1/2 (Way 2: cancel 4s & 5s -> 3/6 = 1.. Keys: [.] 4 [.] 5 [ENTER] [.] 5 [.] 6 [×] [.] 3 [..

PRACTICE B (p.15): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 2/5 of the garden Keys: [.] 2 [.] 3 [ENTER] [.] 3 [.] 5 [×]	**B2:** 2/5 of a full battery Keys: [.] 4 [.] 5 [ENTER] [.] 1 [.] 2 [×]
B3: 4 3/8 cups (35/8) Keys: 1 [.] 3 [.] 4 [ENTER] 2 [.] 1 [.] 2 [×]	**B4:** 3 in² Keys: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]
B5: 4 miles Keys: 3 [.] 1 [.] 5 [ENTER] 1 [.] 1 [.] 4 [×]	**B6:** 2/3 yd² Keys: [.] 3 [.] 4 [ENTER] [.] 8 [.] 9 [×]
B7: 10 gallons Keys: 24 [ENTER] [.] 5 [.] 8 [×] [.] 2 [.] 3 [×]	**B8:** 1/8 in³ Keys: [.] 1 [.] 2 [g][x²]
B9: 8 ft² Keys: 3 [.] 1 [.] 2 [ENTER] 2 [.] 2 [.] 7 [×]	**B10:** 1/12 of the day (2 hours) Keys: [.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [×]

SEC 2.3 (pp.16-17): DIVIDING FRACTIONS AND MIXED NUMBERS [FRAC-01]

Methods: Method 1: Common-Denominator Measurement ('How Many Fit?') | Method 2: Multiply..

PRACTICE A (p.16): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 2 Keys: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [÷]	**A2:** 4 Keys: [.] 2 [.] 3 [ENTER] [.] 1 [.] 6 [÷]
A3: 2 Keys: [.] 5 [.] 6 [ENTER] [.] 5 [.] 12 [f][1/x] [×]	**A4:** 8 Keys: 6 [ENTER] [.] 3 [.] 4 [÷]
A5: 1 7/8 (15/8) Keys: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [÷]	**A6:** 2/5 Keys: [.] 4 [.] 5 [ENTER] 2 [÷]
A7: 6 Keys: 4 [.] 1 [.] 2 [ENTER] [.] 3 [.] 4 [÷]	**A8:** 3 Keys: 5 [.] 1 [.] 4 [ENTER] 1 [.] 3 [.] 4 [÷]
A9: 9 shelves (Way 2: 27 fourths ÷ 3 four.. Keys: 6 [.] 3 [.] 4 [ENTER] [.] 3 [.] 4 [÷]	**A10:** 6 glasses (Way 2: 30/8 ÷ 5/8 = 6) Keys: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

PRACTICE B (p.17): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 6 slices Keys: [.] 3 [.] 4 [ENTER] [.] 1 [.] 8 [÷]	**B2:** 12 snack bags Keys: 4 [.] 1 [.] 2 [ENTER] [.] 3 [.] 8 [÷]
B3: 6 runners Keys: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]	**B4:** 7 segments Keys: 5 [.] 1 [.] 4 [ENTER] [.] 3 [.] 4 [÷]
B5: 40 layers Keys: 6 [ENTER] [.] 3 [.] 20 [÷]	**B6:** 6 spaces Keys: 7 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [÷]
B7: 2 meters Keys: 4 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [÷]	**B8:** 8 cups Keys: 2 [.] 2 [.] 3 [ENTER] [.] 1 [.] 3 [÷]
B9: 5/12 (12/5 × 5/12 = 1) Keys: 2 [.] 2 [.] 5 [f][1/x]	**B10:** 12 pieces Keys: 10 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 2 [-] [...

SEC 2.4 (pp.18-19): ADDING, SUBTRACTING, AND MULTIPLYING DECIMALS [PLACE-01]

Methods: Method 1: Vertical Decimal Alignment / Place Value | Method 2: Inverse Check &..

PRACTICE A (p.18): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: Whole = 27, Decimal = 0.64 Keys: 27.64 [f][IP] / [g][FP]	**A2:** 17.55 Keys: 12.8 [ENTER] 4.75 [+]
A3: 11.35 Keys: 20 [ENTER] 8.65 [-]	**A4:** 2.7 Keys: 4.5 [ENTER] 0.6 [×]
A5: 18.74 Keys: 25.4 [ENTER] 9.86 [-] 3.2 [+]	**A6:** 8.1 Keys: 3.6 [ENTER] 2.25 [×]
A7: 0.06 (3/50) Keys: 0.15 [ENTER] 0.4 [×] [FDISP]	**A8:** 1.35 m² Keys: 1.8 [ENTER] 0.75 [×]
A9: $6.65 (Way 2: 20 - 8.75 - 4.60 = 6.65) Keys: 20 [ENTER] 8.75 [ENTER] 4.60 [+] [-]	**A10:** 5.7 or 5 7/10 (Way 2: 19/4 × 6/5 = 57.. Keys: 4.75 [ENTER] 1.2 [×]

PRACTICE B (p.19): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: $41.24 Keys: 18.75 [ENTER] 22.49 [+]	**B2:** $6.35 Keys: 20 [ENTER] 13.65 [-]
B3: 0.55 seconds faster Keys: 13.42 [ENTER] 12.87 [-]	**B4:** $7.80 Keys: 2.40 [ENTER] 3.25 [×]
B5: 1.19 cm² Keys: 1.4 [ENTER] 0.85 [×]	**B6:** $41.40 Keys: 11.5 [ENTER] 3.60 [×]
B7: 30.35 meters Keys: 24.8 [ENTER] 6.35 [-] 11.9 [+]	**B8:** $35.75 Keys: 2 [ENTER] 14.50 [×] 6.75 [+]
B9: 2.5 mm gap Keys: 170.5 [ENTER] 4 [ENTER] 42 [×] [-]	**B10:** $35.00 profit Keys: 3.20 [ENTER] 1.80 [-] 25 [×]

SEC 2.5 (pp.20-21): DIVIDING WHOLE NUMBERS AND DECIMALS [FRAC-01]

Methods: Method 1: Power-of-10 Decimal Shift | Method 2: Fraction / Mixed-Number Convers..

PRACTICE A (p.20): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 4 R 7, 4.875, 4 7/8 Keys: 39 [ENTER] 8 [÷] [FDISP]	**A2:** 2.3 Keys: 18.4 [ENTER] 8 [÷]
A3: 45 Keys: 11.25 [ENTER] 0.25 [÷]	**A4:** 8 Keys: 7.2 [ENTER] 0.9 [÷]
A5: 130 Keys: 15.6 [ENTER] 0.12 [÷]	**A6:** 6.3 (6 3/10) Keys: 9.45 [ENTER] 1.5 [÷] [FDISP]
A7: 12 Keys: 0.84 [ENTER] 0.07 [÷]	**A8:** 24 (0.2 fits 5 times per 1) Keys: 4.8 [ENTER] 0.2 [÷]
A9: 6 laps (Way 2: 4 1/2 ÷ 3/4 = 6) Keys: 4.5 [ENTER] 0.75 [÷]	**A10:** $1.25/bottle; 16 bottles Keys: 7.50 [ENTER] 6 [÷] -> 1.25; 20 [ENTER] 1.25..

PRACTICE B (p.21): MODELING REAL LIFE APPLICATIONS (#1-#10)

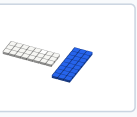
B1: 4 R 3 = 4 3/8 = 35/8 = 4.375 Keys: 35 [ENTER] 8 [÷] [FDISP]	**B2:** 16 quarters Keys: 28 [ENTER] 1.75 [÷]
B3: $8.65 each Keys: 34.60 [ENTER] 4 [÷]	**B4:** 72 layers Keys: 14.4 [ENTER] 0.20 [÷]
B5: 15 bags Keys: 5.25 [ENTER] 0.35 [÷]	**B6:** 6 miles per hour Keys: 4.5 [ENTER] 0.75 [÷]
B7: 18 sections Keys: 22.5 [ENTER] 1.25 [÷]	**B8:** 30 cups Keys: 13.5 [ENTER] 0.45 [÷]
B9: 7 friends Keys: 87.50 [ENTER] 12.50 [÷]	**B10:** 6 Keys: 10.5 [ENTER] 1.75 [÷]

Unit 3: Ratios, Rates & Percents — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Fraction Studio (FRAC-01)

Target Misconception: Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.

Hands-On Block Remediation: Use Blue and Gold Counting Tiles as physical Tape Diagram boxes, step through ratio tables rhythmically with [g] [REP], use [x↔y] when unit-rate division is entered upside-down, and use [f] [%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 3.1 (pp.22-23): RATIOS AND TAPE DIAGRAMS [COUNT-01]

Methods: Method 1: Tape Diagram Model (Equal Boxes) | Method 2: GCF Simplification & Tot..

PRACTICE A (p.22): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 2 : 3 Keys: 12 [ENTER] 18 [g][a:b]	**A2:** 3 : 2 Keys: 24 [ENTER] 16 [g][a:b]
A3: 2 : 5 Keys: 4 [ENTER] 10 [g][a:b]	**A4:** 18, 24 (Total = 42) Keys: 3 [ENTER] 6 [×], 4 [ENTER] 6 [×]
A5: 10 blue, 15 yellow Keys: 25 [ENTER] 5 [÷] -> 5/tile	**A6:** 20 dogs Keys: 36 [ENTER] 9 [÷] 5 [×]
A7: 25 girls Keys: 15 [ENTER] 3 [÷] 5 [×]	**A8:** 30 electric (12 acoustic) Keys: 18 [ENTER] 3 [÷] 5 [×]
A9: 48 and 36 gears (Way 2: 4/7 of 84 = 48) Keys: 84 [ENTER] 7 [÷] -> 12/tile	**A10:** 77 coins total (Way 2: 49 gold + 28 s.. Keys: 21 [ENTER] 3 [÷] 11 [×]

PRACTICE B (p.23): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: Part-to-Part = 3:2; Part-to-Whole = 3:5 Keys: 15 [ENTER] 10 [g][a:b]	**B2:** 16 strawberries (12 bananas) Keys: 28 [ENTER] 7 [÷] 4 [×]
B3: 6 cups yellow (21 cups total) Keys: 15 [ENTER] 5 [÷] 2 [×]	**B4:** 35 dogs and 25 cats Keys: 60 [ENTER] 12 [÷] 7 [×]
B5: 42 gold coins (18 silver) Keys: 24 [ENTER] 4 [÷] 7 [×]	**B6:** 3 : 1 Keys: 48 [ENTER] 16 [g][a:b]
B7: 4 : 9 (16 out of 36) Keys: 16 [ENTER] 36 [g][a:b]	**B8:** 16 : 9 Keys: 1920 [ENTER] 1080 [g][a:b]
B9: 150g almonds, 100g cashews, 250g rais.. Keys: 500 [ENTER] 10 [÷]	**B10:** 3 : 4 Keys: 24 [ENTER] 32 [g][a:b]

SEC 3.2 (pp.24-25): USING RATIO TABLES AND DOUBLE NUMBER LINES [COUNT-01]

Methods: Method 1: Repeated Addition / Double Number Line | Method 2: Multiplicative Sca..

PRACTICE A (p.24): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 4:9, 8:18, 12:27, 16:36 Keys: 4 [g][REP] [÷], 9 [g][REP] [÷]	**A2:** 63 Keys: 28 [ENTER] 4 [÷] 9 [×]
A3: 25 Keys: 60 [ENTER] 12 [÷] 5 [×]	**A4:** 60 pages Keys: 8 [ENTER] 2 [÷] 15 [×]
A5: 48 oz milk Keys: 18 [ENTER] 3 [÷] 8 [×]	**A6:** $15 Keys: 12 [ENTER] 4 [÷] 5 [×]
A7: 84 beats Keys: 60 [ENTER] 15 [÷] 21 [×]	**A8:** No! Multiply by 3 -> 9:15 Keys: 7 [ENTER] 9 [g][a:b]
A9: 20 cups (Way 2: 12÷9 × 15 = 20) Keys: 12 [ENTER] 3 [÷] 5 [×]	**A10:** 40 miles (Way 2: 2 mi per 5% × 20) Keys: 14 [ENTER] 7 [÷] 20 [×]

PRACTICE B (p.25): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 45 grams Keys: 15 [ENTER] 3 [×]	**B2:** 56 scoops Keys: 16 [ENTER] 2 [÷] 7 [×]
B3: $15 for 10 notebooks Keys: 12 [ENTER] 4 [÷] 5 [×]	**B4:** 112 beats per minute Keys: 28 [ENTER] 4 [×]
B5: 260 gallons of sap Keys: 40 [ENTER] 6.5 [×]	**B6:** 25 pounds Keys: 150 [ENTER] 6 [÷]
B7: 15 cups of flour Keys: 6 [ENTER] 2 [÷] 5 [×]	**B8:** 75 minutes (1 hr 15 min) Keys: 90 [ENTER] 18 [÷] 15 [×]
B9: 8 inches Keys: 60 [ENTER] 15 [÷] 2 [×]	**B10:** 30 lemons Keys: 9 [ENTER] 3 [÷] 10 [×]

SEC 3.3 (pp.26-27): RATES, UNIT RATES, AND COMPARING RATES [COUNT-01]

Methods: Method 1: Unit Rate (Divide to Get 1 Unit) | Method 2: Equivalent Rate / Common..

PRACTICE A (p.26): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 65 mph Keys: 195 [ENTER] 3 [÷]	**A2:** $9 / hour Keys: 4 [ENTER] 36 [x⇔y] [÷]
A3: $1.30 / lb Keys: 7.80 [ENTER] 6 [÷]	**A4:** 15 g / hr Keys: 90 [ENTER] 6 [÷]
A5: Cereal B ($0.28/oz vs $0.30/oz) Keys: 4.50 [ENTER] 15 [÷]; 5.04 [ENTER] 18 [÷]	**A6:** 10-pack ($0.82 vs $0.85) Keys: 5.10 [ENTER] 6 [÷]; 8.20 [ENTER] 10 [÷]
A7: 300 miles Keys: 210 [ENTER] 3.5 [÷] 5 [×]	**A8:** Pump A (9 gal/min vs 8 gal/min) Keys: 45 [ENTER] 5 [÷]; 64 [ENTER] 8 [÷]
A9: 180 mm (Way 2: 60s is 4×15s -> 4×45=1.. Keys: 45 [ENTER] 15 [÷] 60 [×]	**A10:** Car B (29 mpg vs 28 mpg) Keys: 168 [ENTER] 6 [÷]; 232 [ENTER] 8 [÷]

PRACTICE B (p.27): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 150 miles per hour Keys: 450 [ENTER] 3 [÷]	**B2:** Box B ($0.70/bar vs. $0.75/bar) Keys: 4.50 [ENTER] 6 [÷] 7.00 [ENTER] 10 [÷]
B3: Spool B ($1.95 vs. $2.20 per 100g) Keys: 11 [ENTER] 5 [÷] 19.50 [ENTER] 10 [÷]	**B4:** 15 gal/min; 375 gallons Keys: 180 [ENTER] 12 [÷] 25 [×]
B5: Leo (57 wpm vs. Maya's 55 wpm) Keys: 220 [ENTER] 4 [÷] 285 [ENTER] 5 [÷]	**B6:** Car A (28 mpg vs. 27 mpg) Keys: 336 [ENTER] 12 [÷] 405 [ENTER] 15 [÷]
B7: $9 per dog -> 11 dogs Keys: 54 [ENTER] 6 [÷] 99 [x⇔y] [÷]	**B8:** 12-pack ($0.42 vs. $0.45 each) Keys: 3.60 [ENTER] 8 [÷] 5.04 [ENTER] 12 [÷]
B9: 4 inches per hour Keys: 14 [ENTER] 3.5 [÷]	**B10:** 4.5 m/s -> 90 meters Keys: 27 [ENTER] 6 [÷] 20 [×]

SEC 3.4 (pp.28-29): PERCENTS, FRACTIONS, AND DECIMALS [FRAC-01]

Methods: Method 1: Scale Denominator to 100 | Method 2: Divide Numerator by Denominator..

PRACTICE A (p.28): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 0.6 and 60% Keys: [.] 3 [.] 5 [ENTER] [FDISP]	**A2:** 0.36 and 36% Keys: [.] 9 [.] 25 [ENTER] [FDISP]
A3: 0.45 and 9/20 Keys: 45 [ENTER] 100 [÷] [FDISP]	**A4:** 12.5% and 1/8 Keys: 0.125 [ENTER] [FDISP]
A5: 24 Keys: 80 [ENTER] 30 [f][%]	**A6:** 9 Keys: 60 [ENTER] 15 [f][%]
A7: 84% Keys: 21 [ENTER] 25 [÷]	**A8:** 1.5 and 1 1/2 Keys: 150 [ENTER] 100 [÷] [FDISP]
A9: 60%, 0.61, 5/8 (Way 2: 60% < 61% < 62.. Keys: [.] 5 [.] 8 [FDISP] -> 0.625	**A10:** 1,400 mAh (Way 2: 7/20 × 4000 = 1400) Keys: 4000 [ENTER] 35 [f][%]

PRACTICE B (p.29): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 0.85 = 85% Keys: [.] 17 [.] 20 [FDISP] 100 [×]	**B2:** 0.36 = 36% Keys: [.] 9 [.] 25 [FDISP] 100 [×]
B3: 20 + 40 = 60 points Keys: 50 [ENTER] 40 [f][%] 50 [ENTER] 80 [f][%] [+]	**B4:** 104 students Keys: 160 [ENTER] 65 [f][%]
B5: 0.15 and 3/20 Keys: [.] 15 [.] 100 [g][SIMP]	**B6:** 45 students total Keys: 18 [ENTER] 0.40 [÷]
B7: $9.00 Keys: 60 [ENTER] 15 [f][%]	**B8:** 7/8 = 0.875 = 87.5% Keys: 21 [ENTER] 24 [÷] [FDISP]
B9: $480 goal Keys: 360 [ENTER] 0.75 [÷]	**B10:** 58% < 3/5 (60%) < 0.62 (62%) Keys: [.] 3 [.] 5 [FDISP]

SEC 3.5 (pp.30-31): SOLVING PERCENT PROBLEMS, DISCOUNTS & CONVERSIONS [PLACE-01]

Methods: Method 1: Subtract the Discount from Original | Method 2: Multiply by Remaining..

PRACTICE A (p.30): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: Discount $24; Sale $56 Keys: 80 [ENTER] 30 [f][%] [-]	**A2:** Tip $6; Total $46 Keys: 40 [ENTER] 15 [f][%] [+]
A3: +30% Keys: 50 [ENTER] 65 [g][%CHG]	**A4:** -25% Keys: 80 [ENTER] 60 [g][%CHG]
A5: 25.4 cm Keys: 10 [f][->cm]	**A6:** 20 in Keys: 50.8 [g][->in]
A7: 4.99 kg Keys: 11 [f][->kg]	**A8:** 72 Keys: 18 [ENTER] 0.25 [÷]
A9: $96 (Way 2: 120 × 0.80 = 96) Keys: 120 [ENTER] 20 [f][%] [-]	**A10:** 81.28 cm, 2.27 kg (Way 2: 32×2.54=81... Keys: 32 [f][->cm]; 5 [f][->kg]

PRACTICE B (p.31): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: Discount = $15; Sale Price = $60 Keys: 75 [ENTER] 20 [f][%] [-]	**B2:** Tip = $9.60; Total = $73.60 Keys: 64 [ENTER] 15 [f][%] [+]
B3: Tax = $3.60; Total = $51.60 Keys: 48 [ENTER] 7.5 [f][%] [+]	**B4:** +30% increase Keys: 40 [ENTER] 52 [g][%CHG]
B5: -30% (30% off) Keys: 90 [ENTER] 63 [g][%CHG]	**B6:** 16.09 km Keys: 10 [ENTER] 1.609344 [×]
B7: 44.09 lb Keys: 20 [g][->lb]	**B8:** 3.79 liters Keys: 1 [f][->L]
B9: $80 - $8 = $72 (28% total off) Keys: 100 [ENTER] 20 [f][%] [-] 10 [f][%] [-]	**B10:** 68°F Keys: 20 [g][->°F]

Unit 4: Algebraic Expressions & Properties — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Manipulative Kits: Algebra Boxes & Factoring Stage (ALGB-01) + Counting Tiles (COUNT-01)

Target Misconception: Students panic when letters appear, erroneously combine 3x + 2 into 5x or $x^2 + x$ into x^3, and don't see why 3(x + 4) = 3x + 12.

Hands-On Block Remediation: Show that a variable x is literally a physical green x-rod (42×14mm) whereas 1 is a small yellow unit cube (14×14mm) and x^2 is a large blue flat (42×42mm)—different physical shapes cannot be counted as one pile! Build rectangles on the L-frame stage for the Distributive Property.



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 4.1 (pp.32-33): WRITING AND EVALUATING ALGEBRAIC EXPRESSIONS [ALGB-01]

Methods: Method 1: Algebra Tile Replacement Model | Method 2: Parentheses Substitution &..

PRACTICE A (p.32): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 33 Keys: 6 [ENTER] 7 [×] 9 [-]	**A2:** 54 Keys: 5 [ENTER] 8 [×] 14 [+]
A3: 13 Keys: 40 [ENTER] 3 [ENTER] 9 [×] [-]	**A4:** 47 Keys: 6 [x^y] 11 [+]
A5: 27 Keys: 4 [x^y] 2 [×] 5 [-]	**A6:** 55 Keys: 7 [ENTER] 4 [×] 3 [ENTER] 9 [×] [+]
A7: 8 Keys: 19 [ENTER] 13 [+] 4 [÷]	**A8:** 15.1 Keys: 5 [ENTER] 3.5 [×] 2.4 [-]
A9: $26 (Way 2: 8+4.5+4.5+4.5+4.5 = 26) Keys: 4.50 [ENTER] 4 [×] 8 [+]	**A10:** 150 cm^2 (Way 2: 6 faces × 25 cm^2 = 150) Keys: 5 [x^y] 6 [×]

PRACTICE B (p.33): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: $26 Keys: 5 [ENTER] 4 [×] 6 [+]	**B2:** 14 tickets Keys: 50 [ENTER] 4 [ENTER] 9 [×] [-]
B3: $225 Keys: 5 [$x^y$] 8 [×] 25 [+]	**B4:** 82 points Keys: 10 [ENTER] 7 [×] 3 [ENTER] 4 [×] [+]
B5: 144 feet Keys: 3 [x^y] 16 [×]	**B6:** 96 in^2 Keys: 4 [x^y] 6 [×]
B7: $41 Keys: 7 [ENTER] 3 [×] 4 [ENTER] 5 [×] [+]	**B8:** 78 Keys: 11 [ENTER] 15 [+] 6 [g][TRI]
B9: 111 minutes (1 hr 51 min) Keys: 12 [ENTER] 8 [×] 15 [+]	**B10:** 128 - 75 = 53 Keys: 4 [g][x^y] 2 [×] 5 [x^y] 3 [×] [-]

SEC 4.2 (pp.34-35): SIMPLIFYING EXPRESSIONS BY COMBINING LIKE TERMS [ALGB-01]

Methods: Method 1: Sort & Group Algebra Tiles by Shape | Method 2: Add/Subtract Coeffici..

PRACTICE A (p.34): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 9x + 13 Keys: 7 [ENTER] 2 [+]; 4 [ENTER] 9 [+]	**A2:** 5x + 7 Keys: 8 [ENTER] 3 [-]; 11 [ENTER] 4 [-]
A3: 8y + 6 Keys: 9 [ENTER] 1 [-]	**A4:** 8a + 13 Keys: 12 [ENTER] 4 [-]; 5 [ENTER] 8 [+]
A5: $6x^2 + 8x + 7$ Keys: 4+2=6; 3+5=8	**A6:** $4n^2 + 5n + 10$ Keys: 6-2=4; 9-4=5
A7: 5.3x + 2.7 Keys: 3.5 [ENTER] 1.8 [+]; 4.2 [ENTER] 1.5 [-]	**A8:** 53 != 80; 5x + 3 is already simplified! Keys: 5(10)+3=53 vs 8(10)=80
A9: 6x + 16 (At x=10: both give 76!) Keys: 2+2+1+1=6x; 3+3+5+5=16	**A10:** 9x + 8 (At x=5: 17+17+19 = 53) Keys: 3+2+4=9x; 2+7-1=8

PRACTICE B (p.35): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 9x + 9 Keys: 2 [ENTER] 3 [+] 4 [+] 3 [ENTER] 1 [+] 5 [+]	**B2:** 8x + 12 Keys: 3 [ENTER] 1 [+] 2 [×] 4 [ENTER] 2 [+] 2 [×]
B3: 4x + 14 Keys: 6 [ENTER] 2 [-] 9 [ENTER] 5 [+]	**B4:** $6x^2 + 4x + 11$ Keys: 4 [ENTER] 2 [+] 7 [ENTER] 3 [-] 3 [ENTER] 8..
B5: 13c + 31 Keys: 8 [ENTER] 5 [+] 12 [ENTER] 19 [+]	**B6:** 7x + 19 (coeff = 7, constant = 19) Keys: 9 [ENTER] 2 [-] 14 [ENTER] 5 [+]
B7: (1/2)n + 13 Keys: [.] 3 [.] 8 [ENTER] [.] 1 [.] 8 [+] 4 [ENTE..	**B8:** 4y + 3.5 Keys: 2.4 [ENTER] 1.6 [+] 5.8 [ENTER] 2.3 [-]
B9: 6m + 9 (both equal 39 at m=5) Keys: 10 [ENTER] 4 [-] 5 [×] 15 [ENTER] 6 [-] [+]	**B10:** $3x^2 + 7$ (the 6x and -6x cancel!) Keys: 5 [ENTER] 2 [-] 10 [ENTER] 3 [-]

SEC 4.3 (pp.36-37): THE DISTRIBUTIVE PROPERTY [ALGB-01]

Methods: Method 1: Total Width First a(b + c) | Method 2: Split Area Model ab + ac | ExA..

PRACTICE A (p.36): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 140 + 21 = 161 Keys: 7 [ENTER] 20 [×] 7 [ENTER] 3 [×] [+]	**A2:** 4x + 20 Keys: 4 [ENTER] 5 [×]
A3: 18x + 24 Keys: 6×3=18; 6×4=24	**A4:** 16y - 40 Keys: 8×2=16; 8×5=40
A5: 387 Keys: 9 [ENTER] 40 [×] 9 [ENTER] 3 [×] [+]	**A6:** 17x + 6 Keys: 3×4+5=17; 3×2=6
A7: 12n + 13 Keys: 4×3=12; 20-7=13	**A8:** 5x + 30 (Must multiply 6 by 5 too!) Keys: 5 [ENTER] 6 [×]
A9: 14x + 15 (At x=2: both equal 43!) Keys: 5×2+4=14; 5×3=15	**A10:** 6x + 21; $60 total Keys: 6 [ENTER] 3.50 [×] -> 6x+21

PRACTICE B (p.37): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 400 + 24 = 424 Keys: 8 [ENTER] 50 [×] 8 [ENTER] 3 [×] [+]	**B2:** $60.00 - $0.60 = $59.40 Keys: 6 [ENTER] 10 [×] 6 [ENTER] 0.10 [×] [-]
B3: 7x + 56 Keys: 7 [ENTER] 8 [×]	**B4:** 18x + 63 Keys: 9 [ENTER] 2 [×] 9 [ENTER] 7 [×]
B5: 4t - 12 ($48 at t = $15) Keys: 4 [ENTER] 15 [×] 4 [ENTER] 3 [×] [-]	**B6:** 14x + 15 Keys: 3 [ENTER] 4 [×] 2 [+] 3 [ENTER] 5 [×]
B7: 13x + 22 Keys: 4 [ENTER] 2 [×] 5 [+] 4 [ENTER] 3 [×] 5 [EN..	**B8:** 3x + 5 Keys: 6 [ENTER] 2 [÷] 10 [ENTER] 2 [÷]
B9: 6x + 16 Keys: 2 [ENTER] 2 [×] 2 [+] 2 [ENTER] 5 [×] 2 [EN..	**B10:** 180 Keys: 12 [ENTER] 15 [×]

SEC 4.4 (pp.38-39): FACTORING EXPRESSIONS USING THE GCF [ALGB-01]

Methods: Method 1: Divide Out GCF Using Prime Factors | Method 2: Reverse Area Model & D..

PRACTICE A (p.38): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 4(3 + 5) Keys: 12 [ENTER] 20 [GCF] -> 4	**A2:** 4(2x + 3) Keys: 8 [ENTER] 12 [GCF] -> 4
A3: 5(3y - 5) Keys: 15 [ENTER] 25 [GCF] -> 5	**A4:** 12(2x + 3) Keys: 24 [ENTER] 36 [GCF] -> 12
A5: 16(2n - 3) Keys: 32 [ENTER] 48 [GCF] -> 16	**A6:** 14(3a + 1) Keys: 42 [ENTER] 14 [GCF] -> 14
A7: 12(3m - 5) Keys: 36 [ENTER] 60 [GCF] -> 12	**A8:** 12(2x + 3) (4 and 6 still share facto.. Keys: 4 [ENTER] 6 [GCF] -> 2 left!
A9: Width = 2x + 3 (Way 2: 14(2x+3)=28x+42) Keys: 28 [ENTER] 42 [GCF] -> 14	**A10:** Side = 9x + 12 (4(9x + 12)) Keys: 36 [ENTER] 4 [÷]; 48 [ENTER] 4 [÷]

PRACTICE B (p.39): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 8(2x + 3) -> 8 bags Keys: 16 [ENTER] 24 [GCF]	**B2:** 7(3x + 5) -> length = 3x + 5 Keys: 21 [ENTER] 35 [GCF]
B3: 4(5x + 9) -> side = 5x + 9 Keys: 20 [ENTER] 36 [GCF]	**B4:** 3(6y - 9) or 9(2y - 3); side = 6y - 9 Keys: 18 [ENTER] 27 [GCF]
B5: 8(5x + 8) Keys: 40 [ENTER] 64 [GCF]	**B6:** 18(2s + 3) -> 18 player kits Keys: 36 [ENTER] 54 [GCF]
B7: 6(2x + 3y + 5) Keys: 12 [ENTER] 18 [GCF] 30 [GCF]	**B8:** 7(5d - 2) Keys: 35 [ENTER] 14 [GCF]
B9: 24x + 36 = 12(2x + 3) Keys: 10 [ENTER] 14 [+] 18 [ENTER] 18 [+] [GCF]	**B10:** 9(3x - 2); at x=2 both equal 36 Keys: 27 [ENTER] 18 [GCF]

Unit 5: Areas of Polygons & Elapsed Time — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Planetary Geared Clock (TIME-01)

Target Misconception: Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.

Hands-On Block Remediation: Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key ([g][7]) which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Planetary Clock (TIME-01) + [CLK] key for base-60 time.

QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!



SEC 5.1 (pp.40-41): AREAS OF PARALLELOGRAMS AND RECTANGLES [COUNT-01]

Methods: Method 1: Cut-and-Slide into a Rectangle | Method 2: Area Formula A = bh & Inve..

PRACTICE A (p.40): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 135 m² (Ignore 11m!)
Keys: 15 [ENTER] 9 [f][RECT]

A2: 216 ft²
Keys: 18 [ENTER] 12 [f][RECT]

A3: 54.4 m²
Keys: 8.5 [ENTER] 6.4 [f][RECT]

A4: 169 cm²
Keys: 13 [x²]

A5: 26 in²
Keys: 6 [.] 1 [.] 2 [ENTER] 4 [f][RECT]

A6: 9 cm
Keys: 108 [ENTER] 12 [÷]

A7: 18 ft
Keys: 135 [ENTER] 7.5 [÷]

A8: 60 (Sam multiplied slanted 7.5!)
Keys: 10 [ENTER] 6 [f][RECT]

A9: 666 ft² (Way 2: base 36 × 18.5 = 666)
Keys: 9 [ENTER] 18.5 [f][RECT] 4 [×]

A10: 168 ft² (Way 2: 180 - 12 = 168)
Keys: 15 [ENTER] 12 [f][RECT] 4 [ENTER] 3 [f][REC..

PRACTICE B (p.41): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 180 ft² (ignore the 20 ft slant!)
Keys: 10 [ENTER] 18 [f][RECT]

B2: 60 ft
Keys: 10 [ENTER] 20 [+] 2 [×]

B3: 14.4 ft²
Keys: 4.5 [ENTER] 3.2 [f][RECT]

B4: 15 inches
Keys: 135 [ENTER] 9 [÷]

B5: 216 cm²
Keys: 6 [ENTER] 4.5 [f][RECT] 8 [×]

B6: 228 ft²
Keys: 19 [ENTER] 12 [f][RECT]

B7: 2,520 ft²
Keys: 65 [ENTER] 40 [f][RECT] 10 [ENTER] 8 [f][RE..

B8: 22 in²
Keys: 5 [.] 1 [.] 2 [ENTER] 4 [×]

B9: Both equal 144 cm²!
Keys: 16 [ENTER] 9 [f][RECT]

B10: 28 ft² × $4 = $112.00
Keys: 8 [ENTER] 3.5 [f][RECT] 4 [×]

SEC 5.2 (pp.42-43): AREAS OF TRIANGLES [COUNT-01]

Methods: Method 1: Half of a Parallelogram/Rectangle | Method 2: Formula A = ½bh (or ½b..

PRACTICE A (p.42): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 126 ft²
Keys: 14 [ENTER] 18 [g][TRI]

A2: 18 ft² (Ignore 9.85!)
Keys: 9 [ENTER] 4 [g][TRI]

A3: 88 cm²
Keys: 16 [ENTER] 11 [g][TRI]

A4: 52.5 m²
Keys: 12.5 [ENTER] 8.4 [g][TRI]

A5: 13 1/2 in² (13.5)
Keys: 4 [.] 1 [.] 2 [ENTER] 6 [g][TRI]

A6: 9 cm
Keys: 54 [ENTER] 2 [×] 12 [÷]

A7: 12 m
Keys: 84 [ENTER] 2 [×] 14 [÷]

A8: 20 (40 is the full rectangle!)
Keys: 8 [ENTER] 5 [g][TRI]

A9: 210 ft² (Way 2: (300+120)÷2 = 210)
Keys: 12 [ENTER] 25 [g][TRI] 8 [ENTER] 15 [g][TRI..

A10: 8 ft (Way 2: 9 [ENTER] 8 [g][TRI] = 36)
Keys: 36 [ENTER] 2 [×] 9 [÷]

PRACTICE B (p.43): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 150 ft²
Keys: 12 [ENTER] 25 [g][TRI]

B2: 14 ft² (1 side); 28 ft² (both)
Keys: 8 [ENTER] 3.5 [g][TRI]

B3: 141 ft²
Keys: 30 [ENTER] 9.4 [g][TRI]

B4: 36 inches
Keys: 450 [ENTER] 2 [×] 25 [÷]

B5: 6 feet
Keys: 18 [ENTER] 2 [×] 6 [÷]

B6: 10.125 in² (10 1/8 in²)
Keys: 4.5 [ENTER] 4.5 [g][TRI]

B7: 54 ft² (use the 9 & 12 legs!)
Keys: 9 [ENTER] 12 [g][TRI]

B8: 32 in²
Keys: 4 [ENTER] 8 [g][TRI] 2 [×]

B9: 13 in²
Keys: 6 [.] 1 [.] 2 [ENTER] 4 [×] 2 [÷]

B10: 64 ft²
Keys: 5 [ENTER] 6.4 [g][TRI] 4 [×]

SEC 5.3 (pp.44-45): AREAS OF TRAPEZOIDS AND COMPOSITE FIGURES [COUNT-01]

Methods: Method 1: Decompose into Rectangles & Triangles | Method 2: Trapezoid Formula A..

PRACTICE A (p.44): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 72 cm²
Keys: 10 [ENTER] 14 [+] 6 [g][TRI]

A2: 135 ft²
Keys: 18 [ENTER] 12 [+] 9 [g][TRI]

A3: 92
Keys: 10 [ENTER] 6 [f][RECT] [Σ+] 8 [ENTER] 4 [f][RECT] [Σ+] [f][Σx]

A4: 120
Keys: 15 [ENTER] 6 [f][RECT] [Σ+] 10 [ENTER] 6 [g][TRI] [Σ+] [f][Σx]

A5: 29.4 m²
Keys: 8.5 [ENTER] 5.5 [+] 4.2 [g][TRI]

A6: 56 in²
Keys: 10 [ENTER] 8 [f][RECT] [Σ+] 6 [ENTER] 4 [f][RECT] [f][Σ-] [f][Σx]

A7: 10 cm
Keys: 90 [ENTER] 2 [×] 11 [ENTER] 7 [+] [÷]

A8: 88 ft²
Keys: 60 [Σ+] 5 [ENTER] 9 [+] 4 [g][TRI] [Σ+] [f][Σx]

A9: 104 (Way 2: 80 + 24 = 104)
Keys: 10 [ENTER] 16 [+] 8 [g][TRI]

A10: 251 ft²
Keys: 300 [Σ+] 24 [f][Σ-] 25 [f][Σ-] [f][Σx]

PRACTICE B (p.45): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 440 in²
Keys: 16 [ENTER] 28 [+] 20 [g][TRI]

B2: 40 in² (1 side); 160 in² (all 4)
Keys: 4 [ENTER] 6 [+] 8 [g][TRI]

B3: 84 + 50 = 134 ft²
Keys: 14 [ENTER] 6 [f][RECT] 10 [ENTER] 5 [f][REC..

B4: 42 cm² (1 fin); 126 cm² (3 fins)
Keys: 5 [ENTER] 9 [+] 6 [g][TRI]

B5: 90 + 48 = 138 in²
Keys: 15 [ENTER] 6 [f][RECT] 12 [ENTER] 8 [g][TRI..

B6: 1,152 - 36 = 1,116 in²
Keys: 48 [ENTER] 24 [f][RECT] 6 [x²] [-]

B7: h = 8 ft
Keys: 104 [ENTER] 13 [÷]

B8: 280 + 60 = 340 ft²
Keys: 20 [ENTER] 14 [f][RECT] 20 [ENTER] 6 [g][TR..

B9: 416 - 200 = 216 m²
Keys: 26 [ENTER] 16 [f][RECT] 20 [ENTER] 10 [f][R..

B10: 42 cm²
Keys: 4 [ENTER] 8 [+] 3.5 [g][TRI] 2 [×]

SEC 5.4 (pp.46-47): POLYGONS IN THE COORDINATE PLANE & ELAPSED TIME [TIME-01]

Methods: Method 1: Grid Counting / Clock Gear Dialing | Method 2: Coordinate Subtraction..

PRACTICE A (p.46): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 8 units
Keys: 11 [ENTER] 3 [-]

A2: 54 sq units
Keys: 10 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

A3: 3 hr 20 min (3.20)
Keys: [CLK] 1.45 [ENTER] 1.35 [+]

A4: 2.5 hours
Keys: 2.30 [f][->HR]

A5: 4.75 hours
Keys: 4.45 [f][->HR]

A6: 1 hr 45 min (1.45)
Keys: 1.75 [g][->HMS]

A7: 2 hr 35 min (2.35)
Keys: [CLK] 11.25 [ENTER] 8.50 [-]

A8: 24 sq units
Keys: 8 [ENTER] 6 [g][TRI]

A9: $52.00 (Way 2: 3×16 + ¼×16 = 48+4 = 52)
Keys: 3.15 [f][->HR] 16 [×]

A10: P = 42 units (420 yd); A = 108 sq uni..
Keys: 12 [ENTER] 9 [f][RECT]

PRACTICE B (p.47): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: W=10, H=6, P=32, Area=60
Keys: 11 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

B2: Base=8, Height=6, Area=24
Keys: 10 [ENTER] 2 [-] 7 [ENTER] 1 [-] [g][TRI]

B3: 12:15 PM
Keys: [CLK] 8.35 [ENTER] 3.40 [+]

B4: 3 hours 35 minutes
Keys: [CLK] 14.20 [ENTER] 10.45 [-]

B5: 3 hours 50 minutes
Keys: [CLK] 1.52 [ENTER] 1.58 [+]

B6: 3.5 hr × $14 = $49.00
Keys: 3.5 [ENTER] 14 [×]

B7: 2 hours 45 minutes
Keys: 2.75 [g][->HMS]

B8: 4 hours 30 minutes
Keys: [CLK] 1.25 [ENTER] 2.15 [+] 0.50 [+]

B9: (3, 10); Area = 36 (it's a 6×6 square!)
Keys: 9 [ENTER] 3 [-] 10 [ENTER] 4 [-] [f][RECT]

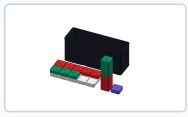
B10: 180 miles
Keys: 2.25 [ENTER] 80 [×]

Unit 6: Integers, Rational Numbers & The Coordinate Plane — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 6 • Manipulative Kits: Ten-Frame Bicolor Counters (BASE-10) + Number Rack (RACK-01)

Target Misconception: Students confuse the subtraction operator [-] with the negative sign [+/-], think -18 > -5 because 18 > 5, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.

Hands-On Block Remediation: Use Yellow (+1) and Red (-1) Zero-Pair counters, test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 6.1 (pp.48-49): POSITIVE AND NEGATIVE INTEGERS & OPPOSITES [BASE-10]

Methods: Method 1: Number Line Distance from 0 | Method 2: Zero-Pair Cancellation & [+/-..

PRACTICE A (p.48): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: -25; Opposite = +25 Keys: 25 [+/-] -> -25; [+/-] -> 25	**A2:** +42; and -42 Keys: 42 [+/-] [+/-] [+/-]
A3: 0 m (Sea level) Keys: 18 [+/-] [ENTER] 18 [+]	**A4:** 0 Keys: 6 [ENTER] 6 [+/-] [+]
A5: 0 ft Keys: 120 [+/-] [ENTER] 120 [+]	**A6:** 0 Keys: 0 [+/-]
A7: +7°F Keys: 12 [+/-] [ENTER] 19 [+]	**A8:** 0 (Even par) Keys: 3 [+/-] [ENTER] 3 [+]
A9: Floor 3 (Way 2: (14+9) - 20 = 3) Keys: 14 [ENTER] 20 [+/-] [+] 9 [+]	**A10:** 0 XP (Break even! 40+25 = 65) Keys: 40 [ENTER] 65 [+/-] [+] 25 [+]

PRACTICE B (p.49): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 0 ft (Sea level; zero pair / opposites) Keys: 120 [+/-] [ENTER] 120 [+]	**B2:** -282 ft; opposite is +282 ft Keys: 282 [+/-]
B3: +19 yards Keys: 12 [ENTER] 8 [+/-] [+] 15 [+]	**B4:** -10°C (10°C below zero) Keys: 15 [ENTER] 25 [-]
B5: 18 zero pairs cancel; +7 remains Keys: 18 [+/-] [ENTER] 25 [+]	**B6:** -5 (5 under par) Keys: 4 [+/-] [ENTER] 2 [+] 3 [+/-] [+]
B7: -x = +35; -(-x) = -35 Keys: 35 [+/-] [+/-]	**B8:** +$10 Keys: 25 [ENTER] 25 [-] 10 [+]
B9: +35 meters Keys: 40 [+/-] [ENTER] 95 [+] 20 [-]	**B10:** +23 (opposite of -23) Keys: 14 [+/-] [ENTER] 9 [+/-] [+] [+/-]

SEC 6.2 (pp.50-51): COMPARING AND ORDERING INTEGERS & RATIONAL NUMBERS [RACK-01]

Methods: Method 1: Horizontal/Vertical Number Line Position | Method 2: Common Decimal o..

PRACTICE A (p.50): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: -5°C (-12 < -5) Keys: 12 [+/-] [ENTER] 5 [+/-] [g][<]	**A2:** -9 < -2 Keys: 9 [+/-] [ENTER] 2 [+/-] [g][<]
A3: -15 < 0 Keys: 15 [+/-] [ENTER] 0 [g][<]	**A4:** -10, -7, -2, 0, 4 Keys: Left-to-right on number line
A5: -3/4 < -0.6 (-0.6 is greater) Keys: 0.75 [+/-] [ENTER] 0.6 [+/-] [g][<]	**A6:** -2 1/2 < -2.3 Keys: 2.5 [+/-] [ENTER] 2.3 [+/-] [g][<]
A7: -5/8 > -7/8 Keys: [.] 5 [.] 8 [+/-] [ENTER] [.] 7 [.] 8 [+/-]..	**A8:** -4.1 is closer to 0 (further right) Keys: 4.5 [+/-] [ENTER] 4.1 [+/-] [g][<]
A9: -1 3/4, -1.5, -0.5, 1 1/4 Keys: -1.75, -1.5, -0.5, 1.25	**A10:** Deepest: -15.2 m; Shallowest: -14 1/2 m Keys: -15.2 < -14.8 < -14.5

PRACTICE B (p.51): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: -11°C > -18°C (Tuesday was warmer) Keys: 11 [+/-] [ENTER] 18 [+/-] [f][>]	**B2:** Alpha (-2.4) is deeper; -2.15 > -2.4 Keys: 2.15 [+/-] [ENTER] 2.4 [+/-] [f][>]
B3: -4/6 > -5/6, so -2/3 > -5/6 Keys: [.] 2 [.] 3 [+/-] [ENTER] [.] 5 [.] 6 [+/-]..	**B4:** -9, -6, -1, +2 Keys: 9 [+/-] [ENTER] 6 [+/-] [g][<]
B5: -89°C, -39°C, -6°C, 0°C Keys: 89 [+/-] [ENTER] 39 [+/-] [g][<]	**B6:** -3.1 > -3 1/4 (-3.25); turtle is higher Keys: 3.25 [+/-] [ENTER] 3.1 [+/-] [g][<]
B7: -1.08% > -1.8% Keys: 1.08 [+/-] [ENTER] 1.8 [+/-] [f][>]	**B8:** No integer; decimal e.g. -3.5 Keys: 3.5 [+/-]
B9: -2, -7/4 (-1.75), -1.6, 3/2 (1.5) Keys: [.] 7 [.] 4 [+/-] [FDISP]	**B10:** a (-10) is further from 0 Keys: 10 [+/-] [g][|x|]

SEC 6.3 (pp.52-53): ABSOLUTE VALUE AND MAGNITUDE [RACK-01]

Methods: Method 1: Count Steps from Zero on Number Line | Method 2: Evaluate |x| Bars Fi..

PRACTICE A (p.52): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 27 Keys: 27 [+/-] [g][|x|]	**A2:** 40 Keys: 27 [+/-] [g][|x|] [ENTER] 13 [+]
A3: 8.0 Keys: 4.8 [+/-] [g][|x|] [ENTER] 3.2 [+/-] [g][|x..	**A4:** 16 Keys: 35 [+/-] [g][|x|] [ENTER] 19 [+/-] [g][|x|]..
A5: |-12| > |9| (12 > 9) Keys: 12 > 9	**A6:** -24 Keys: 24 [+/-] [g][|x|] [+/-]
A7: 32 Keys: 50 [ENTER] 18 [+/-] [g][|x|] [-]	**A8:** -|-9|, -6, 0, |-3|, |-8| Keys: Values: 8, -6, 3, 0, -9
A9: Seagull is closer by 7 ft (25 < 32) Keys: 32 [+/-] [g][|x|] [ENTER] 25 [-]	**A10:** Greater debt: -$45 ($45); Greater bal.. Keys: |-45|=45 > |-28|=28

PRACTICE B (p.53): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: |-28| = 28 ft, |28| = 28 ft (Equal di.. Keys: 28 [+/-] [g][|x|]	**B2:** -$40 is greater balance; Account A ha.. Keys: 65 [+/-] [g][|x|]
B3: 18 + 14 = 32 meters flown Keys: 18 [+/-] [g][|x|] 14 [g][|x|] [+]	**B4:** |-35| = 35; -|-35| = -35 (outside min.. Keys: 35 [+/-] [g][|x|] [+/-]
B5: -1.5 cm (distance |-1.5| = 1.5 cm is.. Keys: 1.5 [+/-] [g][|x|]	**B6:** 42 ÷ 6 + 11 = 18 Keys: 42 [+/-] [g][|x|] 6 [+/-] [g][|x|] [+] 11 [..
B7: 165 feet of cable Keys: 25 [g][|x|] 140 [+/-] [g][|x|] [+]	**B8:** x = 19 and x = -19 Keys: 19 [+/-] [g][|x|]
B9: -|-9| (-9), -6, |5| (5), |-8| (8) Keys: 9 [+/-] [g][|x|] [+/-]	**B10:** 4 5/8 Keys: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+] 1 [..

SEC 6.4 (pp.54-55): THE FOUR-QUADRANT COORDINATE PLANE & REFLECTIONS [COUNT-01]

Methods: Method 1: Plot & Fold Across the Axis | Method 2: Coordinate Sign Rules (Flip y..

PRACTICE A (p.54): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: Quadrant IV Keys: (+, -) -> Right, Down	**A2:** Quadrant III Keys: (-, -) -> Left, Down
A3: Quadrant II Keys: (-, +) -> Left, Up	**A4:** (-6, 9) Keys: 6 [+/-] -> -6
A5: (6, -9) Keys: 9 [+/-] -> -9	**A6:** (-3, 7) Keys: 7 [+/-] [+/-] -> +7
A7: (8, -4) Keys: Flip x -> (8,4); flip y -> (8,-4)	**A8:** (0,-9) on y-axis; (-7,0) on x-axis Keys: 0 sideways vs 0 up/down
A9: (-3, 3) in Quadrant II Keys: 4 [ENTER] 7 [-]; 5 [+/-] [ENTER] 8 [+]	**A10:** D(-4, -5) in Quadrant III Keys: Match x of A (-4) & y of C (-5)

PRACTICE B (p.55): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: Cave: II, Lighthouse: I, Shipwreck: I.. Keys: 8 [+/-] [ENTER] 4	**B2:** (0, -7) is on y-axis; (9, 0) is on x-.. Keys: 0 [ENTER] 7 [+/-]
B3: (8, -12) in Quadrant IV Keys: 8 [ENTER] 12 [+/-]	**B4:** (9, 5) in Quadrant I Keys: 9 [+/-] [+/-] [ENTER] 5
B5: Step 1: (-4, 9); Final: (4, 9) in Qua.. Keys: 4 [+/-] [+/-] [ENTER] 9 [+/-] [+/-]	**B6:** (4, -8) in Quadrant IV Keys: 6 [+/-] [ENTER] 10 [+]
B7: P is in Quadrant II; Q is in Quadrant.. Keys: 2.5 [+/-] [ENTER] 4.5	**B8:** y-axis (x-sign flipped, y stayed -3) Keys: 7 [+/-]
B9: (-5, 5), (-5, -5), and (5, -5) Keys: 5 [+/-]	**B10:** (x, y) is in Quad II; (-x, -y) is in.. Keys: 1 [+/-]

SEC 6.5 (pp.56-57): DISTANCES IN THE COORDINATE PLANE [COUNT-01]

Methods: Method 1: Add/Subtract Absolute Values to Axis | Method 2: Universal Difference..

PRACTICE A (p.56): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 19 units Keys: 8 [+/-] [ENTER] 11 [-] [g][|x|]	**A2:** 11 units Keys: 4 [+/-] [ENTER] 15 [+/-] [-] [g][|x|]
A3: 15 units Keys: 9 [ENTER] 6 [+/-] [-] [g][|x|]	**A4:** 9 units Keys: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]
A5: 10 units Keys: 3.5 [+/-] [ENTER] 6.5 [-] [g][|x|]	**A6:** 8 units Keys: 4.5 [+/-] [ENTER] 3.5 [-] [g][|x|]
A7: (4, 3) Keys: 6 [+/-] [ENTER] 10 [+]	**A8:** 13 units (They are on opposite sides.. Keys: 5 [+/-] [ENTER] 8 [-] [g][|x|]
A9: Perimeter = 34, Area = 72 Keys: Width 9, Height 8	**A10:** 20 units = 10 miles Keys: 10 + 10 = 20 units × 0.5

PRACTICE B (p.57): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: |-8| + |11| = 19 units Keys: 8 [+/-] [ENTER] 11 [-] [g][|x|]	**B2:** 9 units = 90 meters Keys: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]
B3: |15| + |-35| = 50 units Keys: 15 [ENTER] 35 [+/-] [-] [g][|x|]	**B4:** W = 10, H = 6, Perimeter = 32 units Keys: 3 [+/-] [ENTER] 7 [-] [g][|x|] 4 [ENTER] 2..
B5: 60 square units Keys: 10 [ENTER] 6 [f][RECT]	**B6:** 7.75 miles Keys: 3.5 [ENTER] 4.25 [+]
B7: 5 3/4 units Keys: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+]	**B8:** 7 + 10 = 17 units Keys: 5 [ENTER] 2 [+] 4 [ENTER] 6 [+] [+]
B9: (11, 4) and (-5, 4) Keys: 3 [ENTER] 8 [+] 3 [ENTER] 8 [-]	**B10:** Base = 10, Height = 8, Area = 40 units² Keys: 4 [ENTER] 6 [+] 7 [ENTER] 1 [+] [g][TRI]

Unit 7: Equations & Inequalities — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 7 • Manipulative Kits: Algebra Boxes & Stage (ALGB-01) + Number Rack (RACK-01)

Target Misconception: Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of x + 14 = 39, and confuse open vs. solid inequality circles.

Hands-On Block Remediation: Model '=' as a Balance Scale using Algebra Boxes & Stage (ALGB-01). Solve 1-step equations with inverse operations on both sides and check the answer with [f] [LASTX], and test inequalities with [f] [≥] and [g] [≤].



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 7.1 (pp.58-59): WRITING EQUATIONS IN ONE VARIABLE & TESTING SOLUTIONS [ALGB-01]

Methods: Method 1: Substitute & Test Both Sides | Method 2: Mental / Balance Scale Unwra..

PRACTICE A (p.58): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)		PRACTICE B (p.59): MODELING REAL LIFE APPLICATIONS (#1-#10)	
A1: x = 8 (5(8)-7 = 33) Keys: 5 [ENTER] 8 [×] 7 [-]	**A2:** y = 15 (15/3 + 4 = 9) Keys: 15 [ENTER] 3 [÷] 4 [+]	**B1:** x = 8 (5·8 - 7 = 33) Keys: 5 [ENTER] 8 [×] 7 [-]	**B2:** 15t + 5 = 65; t = 4 tickets Keys: 15 [ENTER] 4 [×] 5 [+]
A3: x + 14 = 52 (x = 38) Keys: x + 14 = 52	**A4:** 6k = 84 (k = 14) Keys: 6k = 84	**B3:** k - 12 = 29; k = 41 Keys: 29 [ENTER] 12 [+]	**B4:** n / 6 = 14; n = 84 Keys: 14 [ENTER] 6 [×]
A5: w - 9 = 17 (w = 26) Keys: w - 9 = 17	**A6:** m/5 = 13 (m = 65) Keys: m / 5 = 13	**B5:** 3s = 54; s = 18 cm Keys: 54 [ENTER] 3 [÷]	**B6:** m = 15 (100 - 60 = 40) Keys: 100 [ENTER] 4 [ENTER] 15 [×] [-]
A7: 4p + 6 = 54; p = $12 Keys: 4 [ENTER] 12 [×] 6 [+]	**A8:** x - 12 = 20 (x = 32) Keys: x - 12 = 20	**B7:** p = 24 slices (24 ÷ 8 = 3) Keys: 24 [ENTER] 8 [÷]	**B8:** 45 + d = 120; d = $75 Keys: 120 [ENTER] 45 [-]
A9: m = 6 movies (10 + 3(6) = 28) Keys: 28 [ENTER] 10 [-] 3 [÷]	**A10:** b/3 = 14.50; b = $43.50 Keys: 14.50 [ENTER] 3 [×]	**B9:** Yes! Both sides equal 28 Keys: 3 [ENTER] 6 [×] 10 [+] 5 [ENTER] 6 [×] 2 [-]	**B10:** b = 17 years old Keys: 13 [ENTER] 4 [+]

SEC 7.2 (pp.60-61): SOLVING EQUATIONS USING ADDITION OR SUBTRACTION [ALGB-01]

Methods: Method 1: Inverse Operation (Properties of Equality) | Method 2: Part-Part-Whol..

PRACTICE A (p.60): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)		PRACTICE B (p.61): MODELING REAL LIFE APPLICATIONS (#1-#10)	
A1: x = 45 Keys: 73 [ENTER] 28 [-]	**A2:** y = 92 Keys: 58 [ENTER] 34 [+]	**B1:** s = 165 points Keys: 210 [ENTER] 45 [-]	**B2:** p = $85 Keys: 67 [ENTER] 18 [+]
A3: n = 45 Keys: 64 [ENTER] 19 [-]	**A4:** k = 76 Keys: 29 [ENTER] 47 [+]	**B3:** h = 3,750 feet Keys: 5200 [ENTER] 1450 [-]	**B4:** f = 250.0 grams Keys: 185.2 [ENTER] 64.8 [+]
A5: w = 54.4 Keys: 34.8 [ENTER] 19.6 [+]	**A6:** p = 16.65 Keys: 25.4 [ENTER] 8.75 [-]	**B5:** t = 13.13 seconds Keys: 12.65 [ENTER] 0.48 [+]	**B6:** m = 4/8 = 1/2 cup Keys: [.] 7 [.] 8 [ENTER] [.] 3 [.] 8 [-]
A7: m = 11/24 Keys: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]	**A8:** a = 4 3/4 Keys: 3 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [+]	**B7:** d = 1 3/4 miles Keys: 4 [.] 1 [.] 2 [ENTER] 2 [.] 3 [.] 4 [-]	**B8:** x = 67° Keys: 180 [ENTER] 65 [ENTER] 48 [+] [-]
A9: $45.25 (Check: 45.25 - 18.50 = 26.75) Keys: 26.75 [ENTER] 18.50 [+]	**A10:** 67.5 ft (Check: 67.5 + 42.5 = 110) Keys: 110 [ENTER] 42.5 [-]	**B9:** d = -75 meters (dived 75 m) Keys: 115 [+/-] [ENTER] 40 [+/-] [-]	**B10:** L = 44 cm -> 1 length = 22 cm Keys: 64 [ENTER] 20 [-] 2 [÷]

SEC 7.3 (pp.62-63): SOLVING EQUATIONS USING MULTIPLICATION OR DIVISION [ALGB-01]

Methods: Method 1: Divide by the Coefficient | Method 2: Multiply by the Reciprocal & Ch..

PRACTICE A (p.62): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)		PRACTICE B (p.63): MODELING REAL LIFE APPLICATIONS (#1-#10)	
A1: x = 17 Keys: 136 [ENTER] 8 [÷]	**A2:** y = 15 Keys: 180 [ENTER] 12 [÷]	**B1:** s = $14 per shirt Keys: 168 [ENTER] 12 [÷]	**B2:** p = 27 slices Keys: 3 [ENTER] 9 [×]
A3: m = 114 Keys: 19 [ENTER] 6 [×]	**A4:** k = 101.5 Keys: 14.5 [ENTER] 7 [×]	**B3:** t = 5 hours Keys: 325 [ENTER] 65 [÷]	**B4:** p = $6.50 per pound Keys: 16.25 [ENTER] 2.5 [÷]
A5: w = 14 Keys: 63 [ENTER] 4.5 [÷]	**A6:** x = 24 Keys: 18 [ENTER] [.] 3 [.] 4 [÷]	**B5:** s = $36 Keys: 9 [ENTER] 4 [×]	**B6:** c = 28 students Keys: 21 [ENTER] [.] 3 [.] 4 [÷]
A7: n = 40 Keys: 16 [ENTER] [.] 2 [.] 5 [÷]	**A8:** p = 4/5 (0.8) Keys: [.] 2 [.] 3 [ENTER] [.] 5 [.] 6 [÷]	**B7:** p = $180 Keys: 150 [ENTER] [.] 5 [.] 6 [÷]	**B8:** w = 8 feet Keys: 108 [ENTER] 13.5 [÷]
A9: 18 minutes (Check: 18 × 1.5 = 27) Keys: 27 [ENTER] 1.5 [÷]	**A10:** 45 students (Way 2: 27 ÷ 3 × 5 = 45) Keys: 27 [ENTER] [.] 3 [.] 5 [÷]	**B9:** h = 48 feet Keys: 3.2 [ENTER] 15 [×]	**B10:** n = 140 Keys: 56 [ENTER] 0.40 [÷]

SEC 7.4 (pp.64-65): WRITING AND GRAPHING INEQUALITIES [RACK-01]

Methods: Method 1: Inverse Operation + Boundary Circle Rule | Method 2: Shaded Test-Poin..

PRACTICE A (p.64): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)		PRACTICE B (p.65): MODELING REAL LIFE APPLICATIONS (#1-#10)	
A1: x > 13 (Open circle, arrow Right) Keys: 21 [ENTER] 8 [-]	**A2:** y ≤ 24 (Solid dot, arrow Left) Keys: 15 [ENTER] 9 [+]	**B1:** Solid (■) at 54; g ≥ 5 inches Keys: 54 [ENTER] 49 [-]	**B2:** p ≤ 7 passengers Keys: 12 [ENTER] 5 [-]
A3: n ≥ 13 (Solid dot, arrow Right) Keys: 78 [ENTER] 6 [÷]	**A4:** m < 36 (Open circle, arrow Left) Keys: 9 [ENTER] 4 [×]	**B3:** x ≥ $12.50 Keys: 35 [ENTER] 22.50 [-]	**B4:** g ≤ 8 games Keys: 28 [ENTER] 3.50 [÷]
A5: h ≥ 48 Keys: 'At least' = ≥	**A6:** w ≤ 2000 Keys: 'At most' = ≤	**B5:** r > 16 (Open dot at 16 -> at least 17.. Keys: 240 [ENTER] 15 [÷]	**B6:** b < 40% (Open circle ■ at 40) Keys: 15 [ENTER] 25 [÷]
A7: Only 6 (At n=5, 23 > 23 is False!) Keys: 4(5)+3=23 (False); 4(6)+3=27 (True)	**A8:** 5 and 6 Keys: 4(5)-3=17 ≥ 17 (1=True)	**B7:** 8 and 10 (at x=8, 30 ≥ 30 is TRUE!) Keys: 5 [ENTER] 8 [×] 10 [-]	**B8:** c ≥ 50 cupcakes Keys: 70 [ENTER] 30 [÷] 2 [÷]
A9: At most 6 books ($2 left over) Keys: 50 [ENTER] 8 [g][÷R]	**A10:** p ≥ 160 (340+160=500 ≥ 500 -> True!) Keys: 500 [ENTER] 340 [-]	**B9:** Solid dot at 20, arrow left (0 ≤ s ≤.. Keys: 20 [ENTER] 20 [g][≤]	**B10:** c ≤ 14 crates Keys: 800 [ENTER] 170 [-] 45 [÷]

SEC 7.5 (pp.66-67): WRITING EQUATIONS IN TWO VARIABLES (Y = KX AND Y = X + B) [COUNT-..

Methods: Method 1: Test Constant Ratio y ÷ x vs. Difference y - x | Method 2: Input-Outp..

PRACTICE A (p.66): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)		PRACTICE B (p.67): MODELING REAL LIFE APPLICATIONS (#1-#10)	
A1: y = 8x; y = 80 Keys: 16 [ENTER] 2 [÷] -> 8; 8×10	**A2:** y = x + 8; y = 28 Keys: 9 [ENTER] 1 [-] -> +8; 20+8	**B1:** y = x + 5 is Additive; y = 8m is Mult.. Keys: 8 [ENTER] 1 [×]	**B2:** h = independent, g = dependent; g = 8.. Keys: 18 [ENTER] 4.5 [×]
A3: y = 7x; y = 84 Keys: 14 [ENTER] 2 [÷] -> 7; 7×12	**A4:** y = x - 4; y = 26 Keys: 30 [ENTER] 4 [-]	**B3:** m = 12, 16, 19 (Additive: difference.. Keys: 8 [ENTER] 4 [+]	**B4:** p = 12 sheets Keys: 180 [ENTER] 15 [÷]
A5: (1,6), (2,12), (3,18), (4,24) Keys: 6 [g][REP] [+] [+] [+]	**A6:** Independent: t; Dependent: d Keys: d depends on t	**B5:** $18, $29, and $44 (Additive rule) Keys: 24 [ENTER] 6 [-]	**B6:** 140 wheel turns Keys: 3.5 [ENTER] 40 [×]
A7: 292.5 miles Keys: 65 [ENTER] 4.5 [×]	**A8:** y = 4x hits (0,0); y = x + 4 hits (0,4) Keys: Test x = 0 in both!	**B7:** y = 9x; at x = 20, y = 180 Keys: 27 [ENTER] 3 [÷] 20 [×]	**B8:** c = $127 Keys: 20 [ENTER] 6 [×] 7 [+]
A9: 8 hours (Way 2: 1h=12.5 -> 2h=25 -> 4.. Keys: 100 [ENTER] 12.50 [÷]	**A10:** c = 15m + 25; $205 for 12 months Keys: 15 [ENTER] 12 [×] 25 [+]	**B9:** Multiplicative y = kx (since k × 0 = 0) Keys: 0 [ENTER] 7 [×]	**B10:** 3 miles away Keys: 0.2 [ENTER] 15 [×]

Unit 8: Surface Area & Volume of 3D Solids — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 8 • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Gridfinity 3x1 Storage Boxes

Target Misconception: Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).

Hands-On Block Remediation: Have students touch the 3 pairs of faces on their physical Gridfinity box while logging each pair into [Σ+] (n=1, 2, 3), use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and unlock [f] [CIRC], [g] [CONE], and [g] [HYP].



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 8.1 (pp.68-69): THREE-DIMENSIONAL FIGURES, NETS & RECTANGULAR PRISMS [COUNT-01]

Methods: Method 1: Unfold into a 6-Face 2D Net | Method 2: Surface Area Formula SA = 2(l..

PRACTICE A (p.68): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 96 cm²
Keys: 4 [x²] 6 [×]

A2: 486 in²
Keys: 9 [x²] 6 [×]

A3: 108 cm²
Keys: 2(24)+2(18)+2(12)

A4: 160 in²
Keys: 2(50)+2(20)+2(10)

A5: 392 ft²
Keys: 2(96)+2(60)+2(40)

A6: 85 m²
Keys: 2(20)+2(12.5)+2(10)

A7: 1 face = 25 cm²; s = 5 cm
Keys: 150 [ENTER] 6 [÷] [f][√x]

A8: 94 (He only found 3 faces; must doubl..
Keys: 47 [ENTER] 2 [×]

A9: 260 cm² (Way 2: Full box 340 - 80 lid..
Keys: 80 + 2(50) + 2(40)

A10: 500 in² needed -> Yes, 50 in² to spare!
Keys: 2(150)+2(60)+2(40)

PRACTICE B (p.69): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 2(160 + 600 + 240) = 2,000 cm²
Keys: 20 [ENTER] 8 [×] 20 [ENTER] 30 [×] [+] 8 [E..

B2: 1 face = 81 in²; Total SA = 486 in²
Keys: 9 [x²] 6 [×]

B3: 288 + 720 + 360 = 1,368 in²
Keys: 24 [ENTER] 12 [×] 24 [ENTER] 15 [×] 2 [×] [..

B4: Yes! SA = 500 in² (100 in² to spare)
Keys: 15 [ENTER] 10 [×] 15 [ENTER] 4 [×] [+] 10 [..

B5: 120 + 192 + 160 = 472 ft²
Keys: 12 [ENTER] 10 [×] 12 [ENTER] 8 [×] 2 [×] [+..

B6: 1 face = 25 cm² -> edge s = 5 cm
Keys: 150 [ENTER] 6 [÷] [f][√x]

B7: 32 + 8 = 40 ft²
Keys: 4 [ENTER] 2 [×] 4 [×] 2 [ENTER] 2 [×] 2 [×]..

B8: 2(24 + 60 + 40) = 248 cm²
Keys: 6 [ENTER] 4 [×] 6 [ENTER] 10 [×] [+] 4 [ENT..

B9: 216 cm² (4 times larger: 2² = 4!)
Keys: 6 [x²] 6 [×] 54 [÷]

B10: 160 cm² (2 glued faces are hidden!)
Keys: 4 [x²] 10 [×]

SEC 8.2 (pp.70-71): SURFACE AREAS OF TRIANGULAR PRISMS AND PYRAMIDS [COUNT-01]

Methods: Method 1: Sum Individual Faces on the 2D Net | Method 2: Base Area + Lateral Pe..

PRACTICE A (p.70): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 96 cm²
Keys: 6 [x²] 6 [ENTER] 5 [g][TRI] 4 [×] [+]

A2: 224 in²
Keys: 8 [x²] 8 [ENTER] 10 [g][TRI] 4 [×] [+]

A3: 360 ft²
Keys: 12 [x²] 12 [ENTER] 9 [g][TRI] 4 [×] [+]

A4: 62.4 cm²
Keys: 6 [ENTER] 5.2 [g][TRI] 4 [×]

A5: 132
Keys: 4 [ENTER] 3 [g][TRI] 2 [×] (12) + 12×10 (120)

A6: 150 ft²
Keys: 8 [ENTER] 3 [g][TRI] 2 [×] (24) + 18×7 (126)

A7: 110.88 cm²
Keys: 8 [ENTER] 6.93 [g][TRI] 4 [×]

A8: Prism: 2 tris + 3 rects; Pyramid: 1 s..
Keys: Count net faces

A9: 152 ft² (Way 2: 24 + (6+5+5)×8 = 152)
Keys: 24 + 48 + 80

A10: 108 ft² (Way 2: 27 + 27 + 27 + 27 = 1..
Keys: 9 [ENTER] 6 [g][TRI] 4 [×]

PRACTICE B (p.71): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 24 + 80 + 48 = 152 ft²
Keys: 6 [ENTER] 4 [g][TRI] 2 [×] 5 [ENTER] 8 [f][..

B2: 144 + 240 = 384 cm²
Keys: 12 [x²] 12 [ENTER] 10 [g][TRI] 4 [×] [+]

B3: 96 ft²
Keys: 8 [ENTER] 6 [g][TRI] 4 [×]

B4: 13.8 + 240 = 253.8 cm²
Keys: 6.9 [ENTER] 2 [×] 4 [ENTER] 20 [f][RECT] 3..

B5: 48 + 120 = 168 in²
Keys: 8 [ENTER] 6 [g][TRI] 2 [×] 8 [ENTER] 6 [+]..

B6: 15.6 cm²
Keys: 3 [ENTER] 2.6 [g][TRI] 4 [×]

B7: 96 + 600 = 696 ft²
Keys: 16 [ENTER] 6 [g][TRI] 2 [×] 10 [ENTER] 30 [..

B8: 15 in² per triangular face (slant hei..
Keys: 96 [ENTER] 36 [-] 4 [÷]

B9: 96 + 480 = 576 cm²
Keys: 12 [ENTER] 8 [g][TRI] 2 [×] 10 [ENTER] 10 [..

B10: Pyramid A = 260 > Pyramid B = 224
Keys: 10 [x²] 10 [ENTER] 8 [g][TRI] 4 [×] [+] 8 [..

SEC 8.3 (pp.72-73): VOLUMES OF RECTANGULAR PRISMS [COUNT-01]

Methods: Method 1: Base Area × Height (V = Bh) | Method 2: Unit-Cube Packing & Fractiona..

PRACTICE A (p.72): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 240 cm³
Keys: 10 [ENTER] 6 [ENTER] 4 [f][PRISM]

A2: 125 in³
Keys: 5 [g][x³]

A3: 720 ft³
Keys: 15 [ENTER] 8 [ENTER] 6 [f][PRISM]

A4: 224 m³
Keys: 32 [ENTER] 7 [×]

A5: 10 1/2 ft³ (10.5)
Keys: 3.5 [ENTER] 1.5 [ENTER] 2 [f][PRISM]

A6: 27 in³
Keys: 4.5 [ENTER] 3 [ENTER] 2 [f][PRISM]

A7: 6 in
Keys: 270 [ENTER] 9 [ENTER] 5 [×] [÷]

A8: 8 mini-cubes per 1 in³
Keys: 2 [g][x³]

A9: 8 in (Check: 9 [ENTER] 5 [ENTER] 8 [f..
Keys: 360 [ENTER] 45 [÷]

A10: 216 half-inch cubes
Keys: 27 [ENTER] 2 [g][x³] [×]

PRACTICE B (p.73): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 500 m³
Keys: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

B2: 48 ft³
Keys: 8 [ENTER] 4 [ENTER] 1.5 [f][PRISM]

B3: 15,625 cm³
Keys: 25 [g][x³]

B4: 65 in³
Keys: 6.5 [ENTER] 4 [ENTER] 2.5 [f][PRISM]

B5: 8 per in³ -> 96 half-inch cubes!
Keys: 12 [ENTER] 8 [×]

B6: 15 inches deep
Keys: 3000 [ENTER] 200 [÷]

B7: 77 ft³
Keys: 5.5 [ENTER] 4 [ENTER] 3.5 [f][PRISM]

B8: 36 ft³
Keys: 12 [ENTER] 9 [×] 3 [÷]

B9: 300 + 180 = 480 cm³
Keys: 10 [ENTER] 10 [ENTER] 3 [f][PRISM] 6 [ENTER..

B10: 4,608 in³ ≈ 19.95 (20) gallons
Keys: 24 [ENTER] 12 [ENTER] 16 [f][PRISM] 231 [÷]

SEC 8.4 (pp.74-75): CIRCLES, CONES, AND THE PYTHAGOREAN THEOREM [COUNT-01]

Methods: Method 1: Exact Formula with π (≈ 3.1416) | Method 2: SC-6 Geometry Keys ([f][CIRC], [g][CONE], [g][HYP]) | ExA..

PRACTICE A (p.74): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 78.54 cm²
Keys: 5 [f][CIRC]

A2: 314.16 in²
Keys: 10 [f][CIRC]

A3: 5
Keys: 3 [ENTER] 4 [g][HYP]

A4: 10 ft
Keys: 6 [ENTER] 8 [g][HYP]

A5: 13 ft
Keys: 12 [ENTER] 5 [g][HYP]

A6: 201.06 m²
Keys: 16 [ENTER] 2 [÷] [f][CIRC]

A7: 113.10 cm³
Keys: 3 [ENTER] 12 [g][CONE]

A8: 188.50 in³
Keys: 6 [ENTER] 5 [g][CONE]

A9: 94.25 cm³ (Both ways match 100%!)
Keys: 3 [ENTER] 10 [g][CONE]

A10: 40-inch screen (Way 2: √(1024+576) =..
Keys: 32 [ENTER] 24 [g][HYP]

PRACTICE B (p.75): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 153.94 ft²
Keys: 7 [f][CIRC]

B2: 1,256.64 m²
Keys: 20 [f][CIRC]

B3: 87.96 inches
Keys: 2 [ENTER] [π] [×] 14 [×]

B4: 25 meters
Keys: 15 [ENTER] 20 [g][HYP]

B5: 20 inches
Keys: 16 [ENTER] 12 [g][HYP]

B6: 150.80 cm³ (48π)
Keys: 4 [ENTER] 9 [g][CONE]

B7: 565.49 in³ (180π)
Keys: 6 [ENTER] 15 [g][CONE]

B8: One 16" = 201.06 in² > Two 10" = 157...
Keys: 8 [f][CIRC] 5 [f][CIRC] 2 [×]

B9: 56.55 ft²
Keys: 6 [f][CIRC] 2 [÷]

B10: 17 feet
Keys: 8 [ENTER] 15 [g][HYP]

Unit 9: Statistical Measures & Data Displays — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Probability Dice Studio (PROB-01)

Target Misconception: Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.

Hands-On Block Remediation: Show that Mean is physically leveling unequal cube towers to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 9.1 (pp.76-77): STATISTICAL QUESTIONS, DOT PLOTS, AND THE MEAN [COUNT-01]

Methods: Method 1: Fair-Share Tower Leveling / Balance Point | Method 2: Sum Divided by Count (x-bar = Σx ÷ n) | ExA: Su..

PRACTICE A (p.76): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: Sum = 20, Mean = 5
Keys: [g][CLΣ] 2 [Σ+] 4 [Σ+] 6 [Σ+] 8 [Σ+] [g][x-bar]

A2: Sum = 70, Mean = 14
Keys: Load 5 nums into [Σ+] -> [g][x-bar]

A3: Mean = 21
Keys: Load into [Σ+] -> [g][x-bar]

A4: Mean = 4.0 g
Keys: Load into [Σ+] -> [g][x-bar]

A5: Sum = 55, Mean = 18.33
Keys: ... 35 [f][Σ-] 25 [Σ+] [g][x-bar]

A6: 13.33 (80 ÷ 6)
Keys: 10 [ENTER] 4 [FRQ] 20 [ENTER] 2 [FRQ] [g][x..

A7: 70°F
Keys: Load 6 temps into [Σ+] -> [g][x-bar]

A8: 60 total blocks
Keys: 5 [ENTER] 12 [×]

A9: 95 (Way 2: (-5)+(+2)+(-2)=-5 -> 90+5=..
Keys: 360 [ENTER] 265 [-]

A10: 17 points
Keys: 90 [ENTER] 29 [+] 7 [÷]

PRACTICE B (p.77): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: Sum = 85; Mean = 17 points
Keys: [g][CLΣ] 14 [Σ+] 18 [Σ+] 12 [Σ+] 22 [Σ+] 19 [Σ+] [g][x-bar]

B2: 42 grams
Keys: [g][CLΣ] 41 [Σ+] 43 [Σ+] 42 [Σ+] 40 [Σ+] 44 [Σ+] 42 [Σ+] [g][x-bar]

B3: 89 ÷ 10 = 8.9
Keys: 3 [ENTER] 8 [×] 5 [ENTER] 9 [×] [+] 2 [ENTE..

B4: 425 - 334 = 91 on Test 5
Keys: 5 [ENTER] 85 [×] 80 [ENTER] 88 [+] 82 [+] 8..

B5: Total = 36,000; Mean = 9,000 steps
Keys: 8200 [ENTER] 9500 [+] 7800 [+] 10500 [+] 4..

B6: 270 ÷ 3 = 90.0 (+7.5 pts higher!)
Keys: 88 [ENTER] 92 [+] 90 [+] 3 [÷]

B7: 12 blocks tall
Keys: 6 [ENTER] 10 [+] 14 [+] 18 [+] 4 [÷]

B8: 24.0 ÷ 5 = 4.8 cm
Keys: 4.2 [ENTER] 5.1 [+] 4.8 [+] 5.5 [+] 4.4 [+]..

B9: Reading time varies from student to s..
Keys: 60 [ENTER] 1 [×]

B10: 260 ÷ 10 = 26 points
Keys: 80 [ENTER] 180 [+] 10 [÷]

SEC 9.2 (pp.78-79): MEASURES OF CENTER: MEDIAN, MODE, AND RANGE [PROB-01]

Methods: Method 1: Order Least-to-Greatest by Hand | Method 2: One-Pass [Σ+] Statistical Registers | ExA: Mean = 7.875,..

PRACTICE A (p.78): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: Median = 7, Range = 7 (9 - 2)
Keys: Load 5 nums -> [f][MED], [RNG]

A2: Median = 14, Mode = 12, Range = 8
Keys: Load 6 nums -> [f][MED], [g][MODE], [RNG]

A3: Mean = 9, Med = 8, Mode = 8, Range = 9
Keys: Load 5 nums into [Σ+]

A4: Min = 19, Max = 51, Range = 32
Keys: Load 5 nums -> [f][MIN], [g][MAX], [RNG]

A5: 87 ((84+90)÷2)
Keys: Load 6 nums -> [f][MED]

A6: Mean=$29.17, Med=$17.50 (Median is be..
Keys: Load 6 nums -> [g][x-bar], [f][MED]

A7: Mean jumps 14 -> 30.8; Median stays 14!
Keys: Test both on [Σ+]!

A8: 95
Keys: 64 [ENTER] 31 [+]

A9: Median = 14, Range = 13 (21 - 8)
Keys: Sorted: 8, 11, 14, 16, 21

A10: Mean=$370k, Med=$220k (Trust Median!)
Keys: Load 5 nums -> [g][x-bar], [f][MED]

PRACTICE B (p.79): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: Median = 62, Mode = 62, Range = 22
Keys: [g][CLΣ] 55 [Σ+] 62 [Σ+] 48 [Σ+] 70 [Σ+] 62 [Σ+] 50 [Σ+] 66 [Σ+]..

B2: Ordered: 11,12,[12,13],14,16 -> Media..
Keys: [g][CLΣ] 11 [Σ+] 13 [Σ+] 12 [Σ+] 14 [Σ+] 12 [Σ+] 16 [Σ+] [f][MED]

B3: Median ($25); the $105 outlier skews..
Keys: [g][CLΣ] 20 [Σ+] 25 [Σ+] 20 [Σ+] 30 [Σ+] 105 [Σ+] [g][x-bar] [f][M..

B4: Modes = 6 and 8 (each appears 3 times)
Keys: 6 [ENTER] 8

B5: Median = 11.8 s, Range = 1.2 s, No mo..
Keys: 12.4 [ENTER] 11.2 [-]

B6: Min = 64, Max = 81, Range = 17°F, Med..
Keys: [g][CLΣ] 72 [Σ+] 68 [Σ+] 75 [Σ+] 81 [Σ+] 64 [Σ+] 70 [Σ+] 74 [Σ+]..

B7: Mean = 11, Median = 11, Mode = 12, Ra..
Keys: [g][CLΣ] 8 [Σ+] 12 [Σ+] 9 [Σ+] 15 [Σ+] 12 [Σ+] 10 [Σ+] [g][x-bar] [..

B8: Example: 8, 8, 10, 12, 15
Keys: 15 [ENTER] 8 [-]

B9: Median increases by +5 (to 85); Range..
Keys: 95 [ENTER] 75 [-]

B10: Mean = 11.75, Median = 11.5, Mode = 1..
Keys: [g][CLΣ] 9 [Σ+] 12 [Σ+] 10 [Σ+] 14 [Σ+] 11 [Σ+] 15 [Σ+] 10 [Σ+] 1..

SEC 9.3 (pp.80-81): QUARTILES, INTERQUARTILE RANGE (IQR) & BOX PLOTS [FRAC-01]

Methods: Method 1: Split Ordered Data into Lower & Upper Halves | Method 2: [1Q], [MED],..

PRACTICE A (p.80): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: Min=10, Q1=15, Med=30, Q3=45, Max=50,..
Keys: Load 5 nums -> [f][1Q], [g][3Q], [IQR]

A2: Q1=6, Med=10, Q3=14, IQR=8
Keys: Load 7 nums -> [f][1Q], [g][3Q], [IQR]

A3: 34
Keys: 58 [ENTER] 24 [-]

A4: 75%
Keys: 3 quarters = 3 × 25%

A5: 68, 73.5, 82, 90, 96; IQR = 16.5
Keys: Load 9 nums into [Σ+]

A6: Q1=46.5, Med=51, Q3=56.5, IQR=10
Keys: Load 8 nums into [Σ+]

A7: Player A (smaller IQR = tighter clust..
Keys: Compare box widths

A8: 90 (Any score > 90 is an outlier)
Keys: 60 [ENTER] 1.5 [ENTER] 20 [×] [+]

A9: IQR = 13 (28 - 15 = 13)
Keys: Load 7 nums -> [IQR]

A10: 100 students (50% of 200)
Keys: 200 [ENTER] 50 [f][%]

PRACTICE B (p.81): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: Min=40, Q1=45, Med=55, Q3=65, Max=70;..
Keys: [g][CLΣ] 40 [Σ+] 45 [Σ+] 50 [Σ+] 55 [Σ+] 60 [Σ+] 65 [Σ+] 70 [Σ+]..

B2: Q1 = 7.5, Med = 10, Q3 = 13, IQR = 5.5
Keys: [g][CLΣ] 6 [Σ+] 7 [Σ+] 8 [Σ+] 9 [Σ+] 11 [Σ+] 12 [Σ+] 14 [Σ+] 15 [Σ..

B3: Q1 = 22.5, Med = 30, Q3 = 42.5, IQR =..
Keys: [g][CLΣ] 30 [Σ+] 15 [Σ+] 45 [Σ+] 25 [Σ+] 35 [Σ+] 20 [Σ+] 50 [Σ+]..

B4: Range = 28 ft; IQR = 13 ft
Keys: 40 [ENTER] 12 [-] 31 [ENTER] 18 [-]

B5: 75% of the trees = 45 trees
Keys: 60 [ENTER] 75 [f][%]

B6: 25% of 60 = 15 trees
Keys: 60 [ENTER] 25 [f][%]

B7: Class A (smaller IQR = tighter middle..
Keys: 19 [ENTER] 8 [-]

B8: 30 + 15 = 45 (anything > 45 is an out..
Keys: 30 [ENTER] 1.5 [ENTER] 10 [×] [+]

B9: Min=18, Q1=22, Med=27, Q3=34, Max=40;..
Keys: [g][CLΣ] 18 [Σ+] 22 [Σ+] 25 [Σ+] 29 [Σ+] 34 [Σ+] 40 [Σ+] |IQR|

B10: Q1 = 12, Q3 = 20, IQR = 8 (unaffected..
Keys: [g][CLΣ] 10 [Σ+] 12 [Σ+] 14 [Σ+] 16 [Σ+] 18 [Σ+] 20 [Σ+] 100 [Σ+..

SEC 9.4 (pp.82-83): MEAN ABSOLUTE DEVIATION (MAD) [COUNT-01]

Methods: Method 1: 3-Step Distance Table (Mean -> |x - x-bar| -> Average) | Method 2: [Σ+] & [g] [MAD] Verification | ExA: M..

PRACTICE A (p.82): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: Mean = 4, MAD = 1.33 (4/3)
Keys: 2 [Σ+] 4 [Σ+] 6 [Σ+] [g][MAD]

A2: Mean = 14, MAD = 2.4
Keys: Load 5 nums -> [g][x-bar], [g][MAD]

A3: Mean = 9, MAD = 2.4
Keys: Load 5 nums -> [g][x-bar], [g][MAD]

A4: Mean = 7, MAD = 0
Keys: All distances from 7 are 0!

A5: Mean = 28, MAD = 4.8
Keys: Load 5 nums -> [g][x-bar], [g][MAD]

A6: Mean = 13.0, MAD = 0.40
Keys: Load 6 decimals -> [g][x-bar], [g][MAD]

A7: Mean = 88, MAD = 3.6
Keys: Load 5 nums -> [g][x-bar], [g][MAD]

A8: Team B (larger MAD = more spread out)
Keys: Compare MADs

A9: Mean = 10, MAD = 2.0
Keys: Mean=10; dists 4+0+1+3=8 ÷ 4

A10: x-bar=9, Med=8, Mode=8, Rng=9, IQR=5...
Keys: 1 entry -> 6 stats!

PRACTICE B (p.83): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: Mean = 210°C; distances 4,2,2,4 -> MA..
Keys: [g][CLΣ] 206 [Σ+] 208 [Σ+] 212 [Σ+] 214 [Σ+] [g][MAD]

B2: Mean = 10; MAD = 1.6 shots
Keys: [g][CLΣ] 7 [Σ+] 9 [Σ+] 10 [Σ+] 11 [Σ+] 13 [Σ+] [g][x-bar] [g][MAD]

B3: Mean = 50g, MAD = 0 (zero variation!)
Keys: 0 [ENTER]

B4: Archer B MAD = 2.0 -> Archer A is mor..
Keys: [g][CLΣ] 6 [Σ+] 8 [Σ+] 10 [Σ+] 12 [Σ+] [g][MAD]

B5: Mean = 85; distances 9,3,0,3,9 -> MAD..
Keys: [g][CLΣ] 76 [Σ+] 82 [Σ+] 85 [Σ+] 88 [Σ+] 94 [Σ+] [g][x-bar] [g][M..

B6: Mean = 15 m, MAD = 1.0 m
Keys: [g][CLΣ] 15 [Σ+] 16 [Σ+] 14 [Σ+] 15 [Σ+] 17 [Σ+] 13 [Σ+] [g][x-ba..

B7: Deviations: 5, 1, 1, 5 -> MAD = 3.0°F
Keys: [g][CLΣ] 80 [Σ+] 84 [Σ+] 86 [Σ+] 90 [Σ+] [g][MAD]

B8: They always cancel to 0! |x - x-bar|..
Keys: 5 [+/-] [ENTER] 5 [+]

B9: Mean = 5 min; MAD = 2.4 min
Keys: [g][CLΣ] 2 [Σ+] 3 [Σ+] 4 [Σ+] 5 [Σ+] 11 [Σ+] [g][x-bar] [g][MAD]

B10: Mean=16, Med=15, Mode=15, Rng=8, IQR=..
Keys: [g][CLΣ] 12 [Σ+] 15 [Σ+] 15 [Σ+] 18 [Σ+] 20 [Σ+] [g][x-bar] [f][ME..

Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability — Practice A &..

Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Manipulative Kits: Ten-Frame Zero-Pair Counters (BASE-10) + Algebra Stage (ALGB-01) + Probability Studio (PROB-01)

Target Misconception: Students guess signs when adding/subtracting negatives, get -25 instead of +25 for (-5)² on algebraic calculators, undo ×3 before +7 in 3x + 7 = 28, and confuse Permutations (order matters) with Combinations (order doesn't).

Hands-On Block Remediation: Use Yellow/Red counters to show removing 4 red debt tokens (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x²] squares the entire register (-5)² = +25 without parentheses! Unwrap 2-step equations in reverse (untie ribbon ±b first, open box ×a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].



QUEST-01 Double-Points Scoring: 1 '+10' token = Method 1 (1st Way); stack a 2nd '+10' token (+20 pts) = Method 2 (2nd Way + SC-6 Check)!

SEC 10.1 (pp.84-85): ADDING AND SUBTRACTING RATIONAL NUMBERS [BASE-10]

Methods: Method 1: Zero-Pair Counters / Number Line Jumps | Method 2: Absolute Value Rul..

PRACTICE A (p.84): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: (a) -6; (b) -24
Keys: 15 [+/-] [ENTER] 9 [+] / 9 [+/-] [+]

A2: -9
Keys: 14 [ENTER] 23 [+/-] [+]

A3: (a) 20; (b) -4
Keys: 12 [ENTER] 8 [+/-] [-] / 12 [+/-]...

A4: -25
Keys: 9 [+/-] [ENTER] 16 [-]

A5: -8.3
Keys: 4.8 [+/-] [ENTER] 3.5 [+/-] [+]

A6: -4.8
Keys: 6.4 [ENTER] 11.2 [-]

A7: +2.7
Keys: 7.5 [+/-] [ENTER] 10.2 [+/-] [-]

A8: -1/8
Keys: [.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-]..

A9: +25°F (Way 2: 18 steps to 0 + 7 steps..
Keys: 7 [ENTER] 18 [+/-] [-]

A10: 2 7/8 (2.875)
Keys: 2.5 [+/-] [ENTER] 4.25 [+] 1.125 [+]

PRACTICE B (p.85): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: +8°F
Keys: 14 [+/-] [ENTER] 22 [+]

B2: -83 meters
Keys: 45 [+/-] [ENTER] 38 [+/-] [+]

B3: 5,000 meters
Keys: 4800 [ENTER] 200 [+/-] [-]

B4: -15 + 25 = +10 points
Keys: 15 [+/-] [ENTER] 25 [+/-] [-]

B5: +$8.75
Keys: 12.50 [+/-] [ENTER] 30 [+] 8.75 [-]

B6: 1/8
Keys: [.] 5 [.] 8 [+/-] [ENTER] [.] 3 [.] 4 [+]

B7: -9/6 = -1 1/2
Keys: [.] 2 [.] 3 [+/-] [ENTER] [.] 5 [.] 6 [-]

B8: -16.0 meters
Keys: 18.4 [+/-] [ENTER] 7.9 [+] 5.5 [-]

B9: -15.0°C change
Keys: 8.5 [+/-] [ENTER] 6.5 [-]

B10: -8
Keys: 14 [+/-] [ENTER] 20 [+/-] [-] 9 [+/-] [+] 5..

SEC 10.2 (pp.86-87): MULTIPLYING AND DIVIDING RATIONAL NUMBERS [BASE-10]

Methods: Method 1: Repeated Signed Addition / Sign-Flip Pattern | Method 2: Count Negati..

PRACTICE A (p.86): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: (a) -56; (b) +56
Keys: 7 [ENTER] 8 [+/-] [×] 7 [+/-]...

A2: (a) -5; (b) +5
Keys: 45 [+/-] [ENTER] 9 [÷] / 9 [+/-] [÷]

A3: +25 (2 negatives = +)
Keys: 5 [+/-] [x²]

A4: -8 (3 negatives = odd = -)
Keys: 2 [+/-] [g][x²]

A5: +8.4
Keys: 3.5 [+/-] [ENTER] 2.4 [+/-] [×]

A6: -12
Keys: 14.4 [+/-] [ENTER] 1.2 [÷]

A7: -3/5 (-0.6)
Keys: [.] 2 [.] 3 [+/-] [ENTER] [.] 9 [.] 10 [×]

A8: +2
Keys: [.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-..

A9: -60 (3 negatives = odd -> negative!)
Keys: 4 [+/-] [ENTER] 5 [+/-] [×] 3 [+/-] [×]

A10: -45 ft/min (Check: -45 × 8 = -360 ft)
Keys: 360 [+/-] [ENTER] 8 [÷]

PRACTICE B (p.87): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: -120 feet
Keys: 8 [ENTER] 15 [+/-] [×]

B2: +40 points (Same signs = Positive)
Keys: 4 [+/-] [ENTER] 10 [+/-] [×]

B3: -4.5°F per hour
Keys: 27 [+/-] [ENTER] 6 [÷]

B4: (-4)² = +16 (2 negs); (-4)³ = -64 (3..
Keys: 4 [+/-] [x²] 4 [+/-] [g][x³]

B5: +2/3
Keys: [.] 3 [.] 4 [+/-] [ENTER] [.] 8 [.] 9 [+/-]..

B6: -2
Keys: [.] 7 [.] 8 [+/-] [ENTER] [.] 7 [.] 16 [÷]

B7: -12 (Odd number of negatives = -)
Keys: 2.5 [+/-] [ENTER] 1.6 [+/-] [×] 3 [+/-] [×]

B8: -3.2 meters per year
Keys: 14.4 [+/-] [ENTER] 4.5 [÷]

B9: +120
Keys: 2 [+/-] [ENTER] 3 [+/-] [×] 4 [+/-] [×] 5 [..

B10: 36 - (-20) + 6 = 62
Keys: 6 [+/-] [x²] 4 [ENTER] 5 [+/-] [×] [-] 18 [..

SEC 10.3 (pp.88-89): SOLVING TWO-STEP EQUATIONS (AX + B = C) [ALGB-01]

Methods: Method 1: 2-Step Inverse Operations | Method 2: Algebra Tile Balance Scale & Su..

PRACTICE A (p.88): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 4x = 36 -> x = 9
Keys: 45 [ENTER] 9 [-] 4 [÷]

A2: 5x = 50 -> x = 10
Keys: 38 [ENTER] 12 [+] 5 [÷]

A3: 6y = 48 -> y = 8
Keys: 59 [ENTER] 11 [-] 6 [÷]

A4: 8n = 64 -> n = 8
Keys: 49 [ENTER] 15 [+] 8 [÷]

A5: m/4 = 9 -> m = 36
Keys: 15 [ENTER] 6 [-] 4 [×]

A6: k/3 = 12 -> k = 36
Keys: 7 [ENTER] 5 [+] 3 [×]

A7: -3x = -12 -> x = 4
Keys: 2 [ENTER] 14 [-] 3 [+/-] [÷]

A8: 2.5w = 15 -> w = 6
Keys: 23 [ENTER] 8 [-] 2.5 [÷]

A9: h = 5 hours (Check: 25(5)+35 = 160)
Keys: 160 [ENTER] 35 [-] 25 [÷]

A10: w = 5 weeks (Check: 15(5)+45 = 120)
Keys: 120 [ENTER] 45 [-] 15 [÷]

PRACTICE B (p.89): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: r = 5 laps
Keys: 38 [ENTER] 8 [-] 6 [÷]

B2: h = 9 hours
Keys: 48 [ENTER] 12 [-] 4 [÷]

B3: w = 7 weeks
Keys: 140 [ENTER] 35 [-] 15 [÷]

B4: b = $14 per book
Keys: 46 [ENTER] 10 [+] 4 [÷]

B5: x = 6 (-4·6 + 18 = -6)
Keys: 6 [+/-] [ENTER] 18 [-] 4 [+/-] [÷]

B6: n = 35
Keys: 16 [ENTER] 9 [-] 5 [×]

B7: m = 60 minutes
Keys: 600 [ENTER] 3000 [-] 40 [+/-] [÷]

B8: x = 24
Keys: 11 [ENTER] 5 [+] [.] 2 [.] 3 [÷]

B9: w = 10 cm
Keys: 56 [ENTER] 36 [-] 2 [÷]

B10: m = 8 miles
Keys: 21.50 [ENTER] 3.50 [-] 2.25 [÷]

SEC 10.4 (pp.90-91): THEORETICAL PROBABILITY, COMBINATIONS & PERMUTATIONS [PROB-01]

Methods: Method 1: Favorable ÷ Total & Complement Rule | Method 2: Fundamental Counting..

PRACTICE A (p.90): KEY IDEAS & TWO-METHOD PRACTICE (#1-#10)

A1: 1/6; and 1/2 (50%)
Keys: 1/6; and 3/6 -> [g][SIMP]

A2: 2/5, 0.4, 40%
Keys: [.] 4 [.] 10 [g][SIMP] [FDISP]

A3: 7/10 (70%)
Keys: (10 - 3) / 10 = 7/10

A4: 24 ways
Keys: 4 [g][x!]

A5: 720 ways
Keys: 6 [g][x!]

A6: 56 teams
Keys: 8 [ENTER] 3 [f][nCr]

A7: 336 podiums
Keys: 8 [ENTER] 3 [g][nPr]

A8: 1/6
Keys: [.] 1 [.] 2 [ENTER] [.] 1 [.] 3 [×]

A9: 120 vs 720 (120 × 3! = 720)
Keys: 10 [ENTER] 3 [f][nCr] / [g][nPr]

A10: 120 ways (Course Complete!)
Keys: 5 [g][x!]; then [f][?]

PRACTICE B (p.91): MODELING REAL LIFE APPLICATIONS (#1-#10)

B1: 4/20 = 1/5 = 0.20 = 20%
Keys: [.] 4 [.] 20 [g][SIMP] [FDISP]

B2: 65% (13/20 = 0.65)
Keys: 100 [ENTER] 35 [-]

B3: P(green) = 1/3; P(not gold) = 11/15
Keys: [.] 5 [.] 15 [g][SIMP]

B4: 120 custom skateboards
Keys: 5 [ENTER] 4 [×] 6 [×]

B5: 5! = 120 orders
Keys: 5 [g][x!]

B6: 210 batting orders
Keys: 7 [ENTER] 3 [g][nPr]

B7: 35 lab groups
Keys: 7 [ENTER] 3 [f][nCr]

B8: 28 fruit combinations
Keys: 8 [ENTER] 2 [f][nCr]

B9: 60 times
Keys: 160 [ENTER] [.] 3 [.] 8 [×]

B10: 12 outcomes; P(H and 6) = 1/12
Keys: 2 [ENTER] 6 [×]